

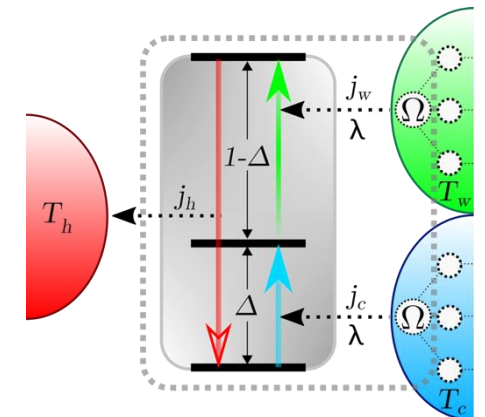
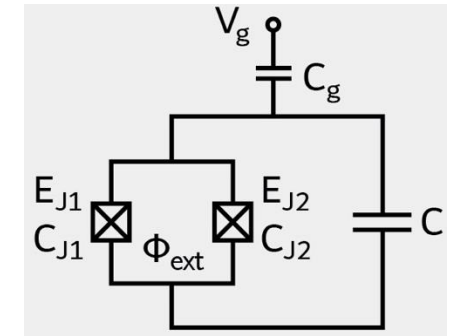
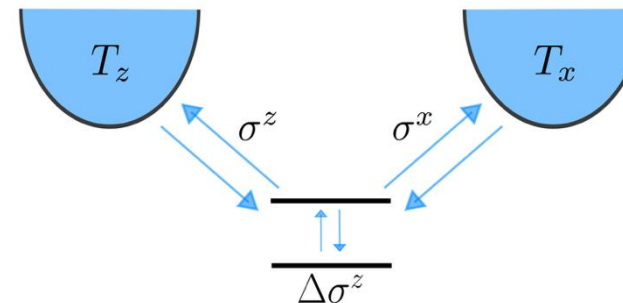
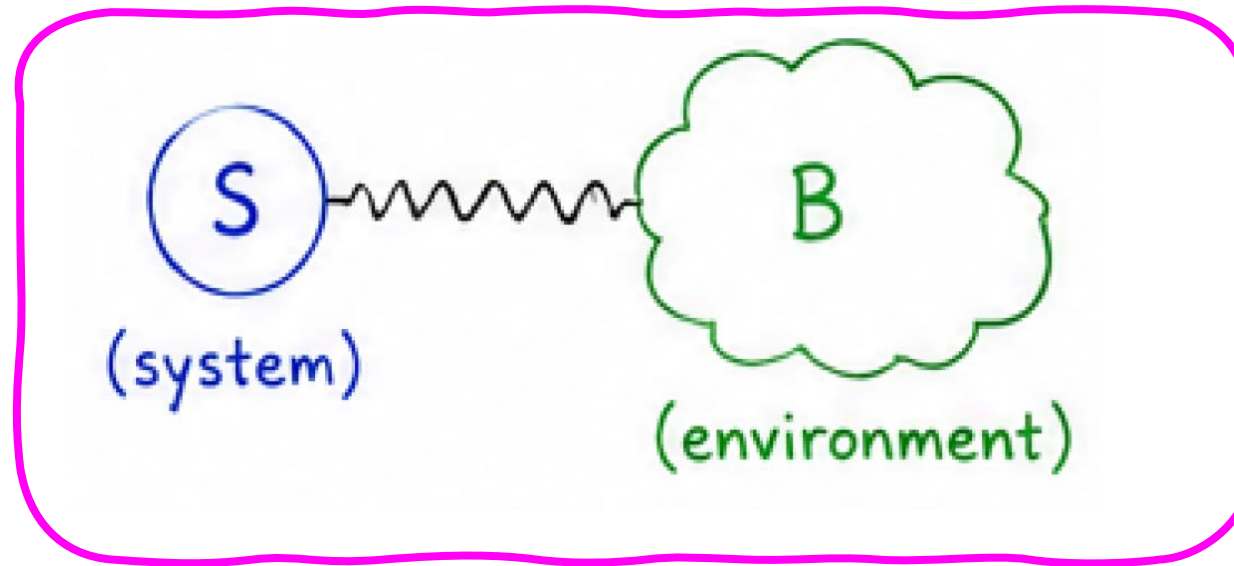
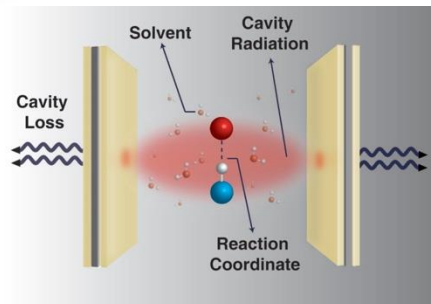
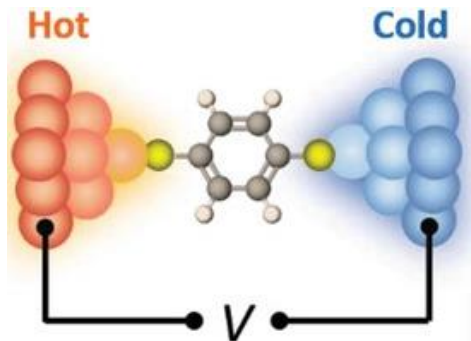
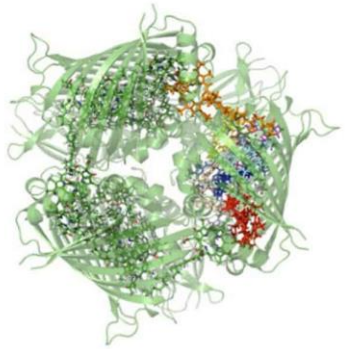
Open quantum systems: Intro

Dvira Segal

Departments of Chemistry and Physics

Director: CQIQC

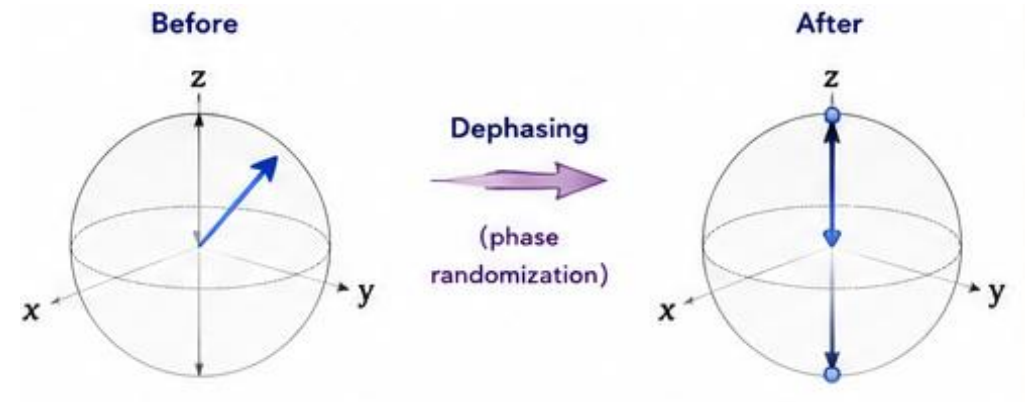
University of Toronto



Outline: Talk #1

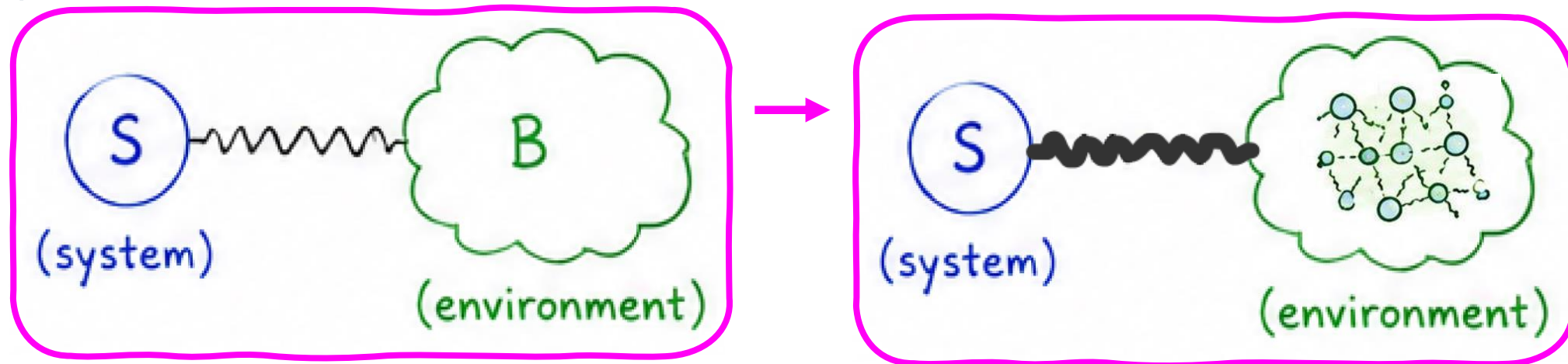
○ Intro

- Open Quantum Systems
- Density matrix, Interaction representation
- Spin Boson model
- T1 and T2 times



○ Derivation of the Quantum Master Equation:

- Redfield Equation
- Lindblad Equation
- Examples



Outline: Today's talk #2

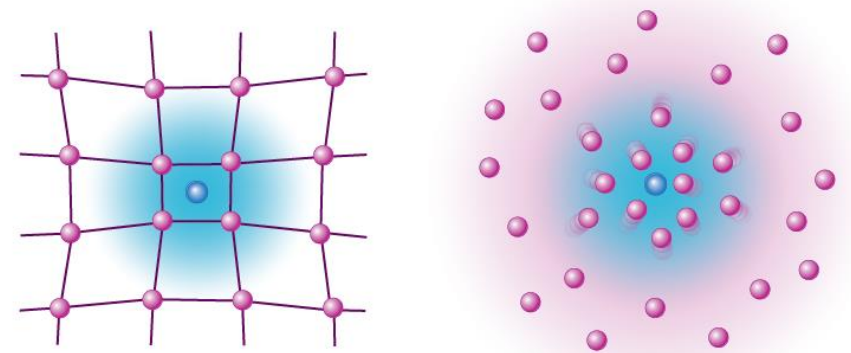
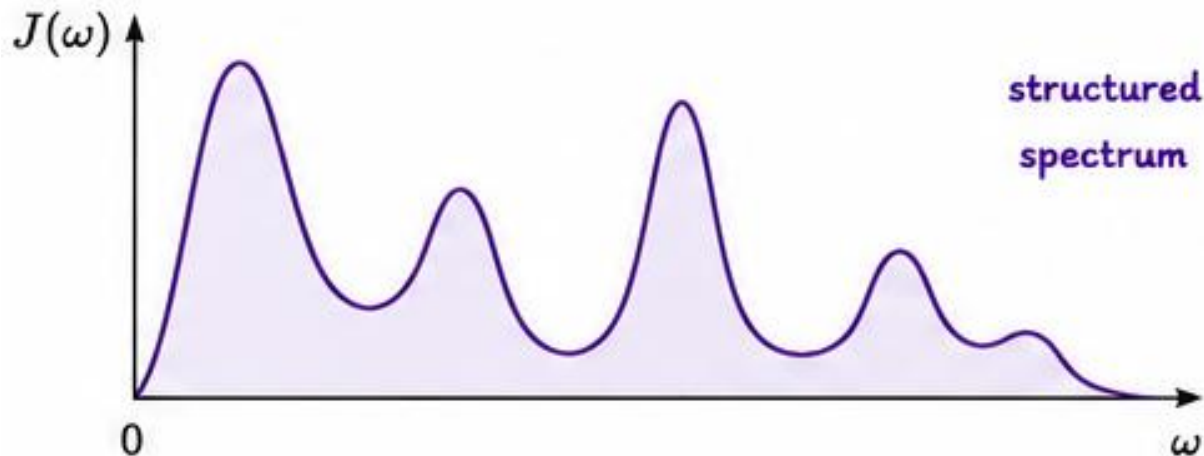
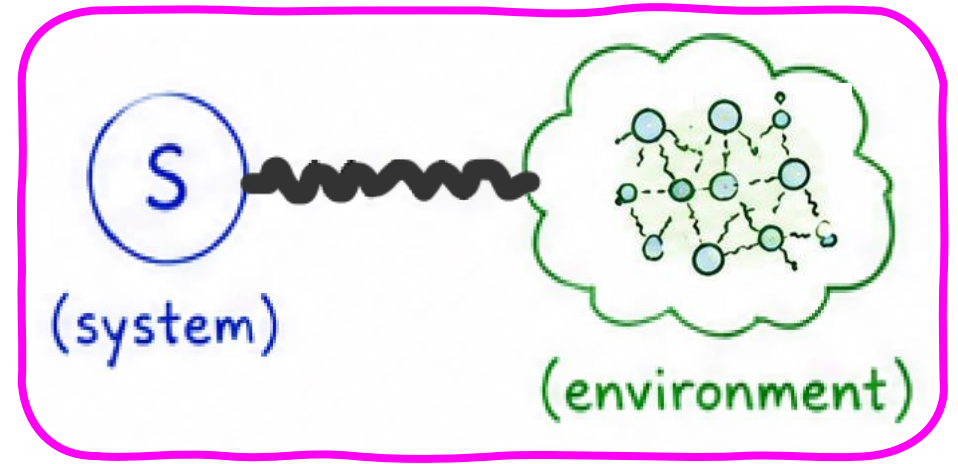
○ Review

- Weak coupling, Markovian approximation
- Quantum master equation

○ Memory effects: Measures, Methods

○ Strong coupling: Polaron mapping

○ Other methods: Repeated interaction, Path integral, HEOM,...



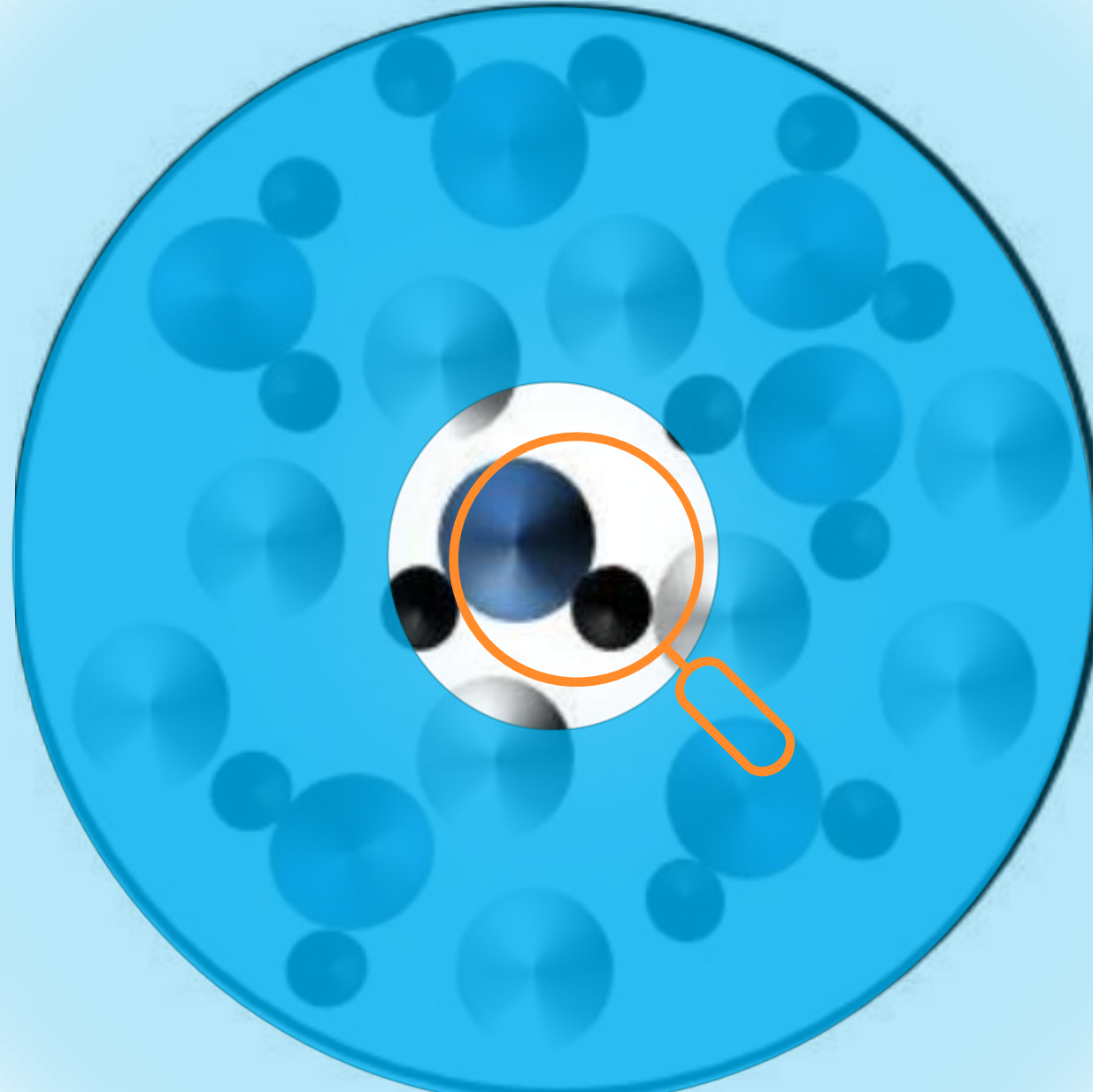
Theory of Open Quantum systems

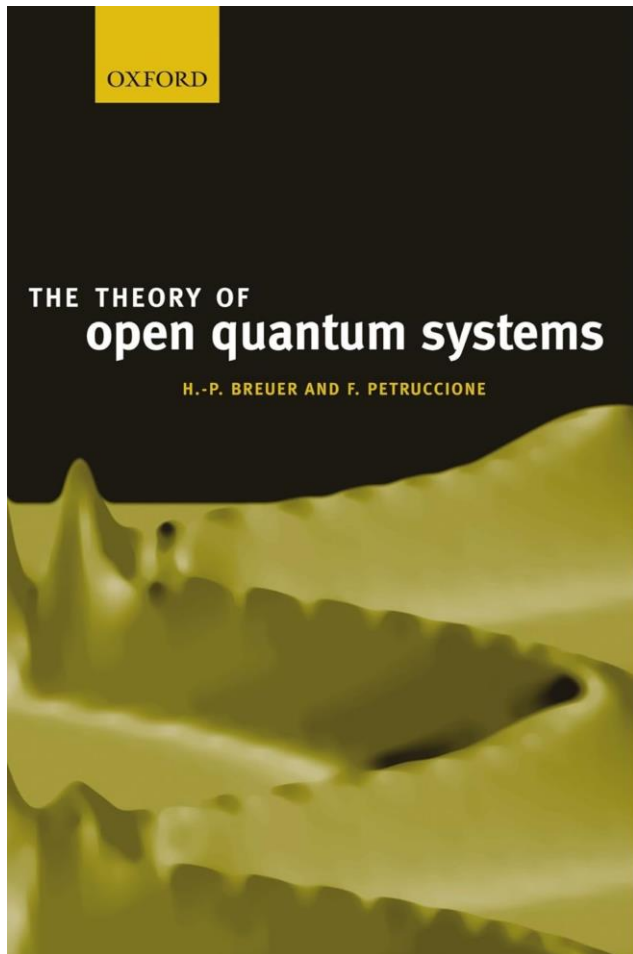
Fundamental and applications

Method 1:

Solve accurately for all degrees of freedom, then focus.

Method 2: Focus-solve accurately for the part that interests me while treating the rest approximately and with less details.

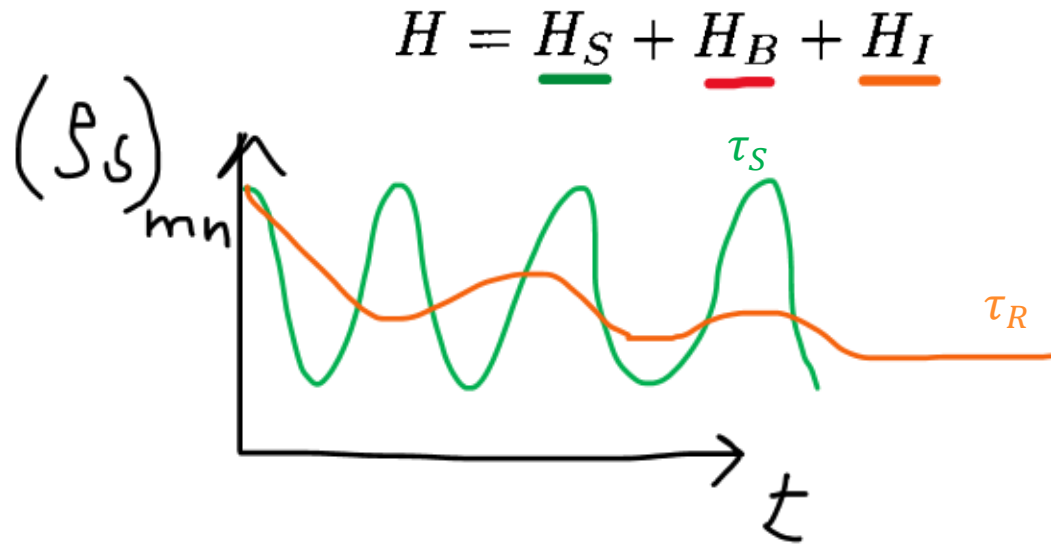




II DENSITY MATRIX THEORY

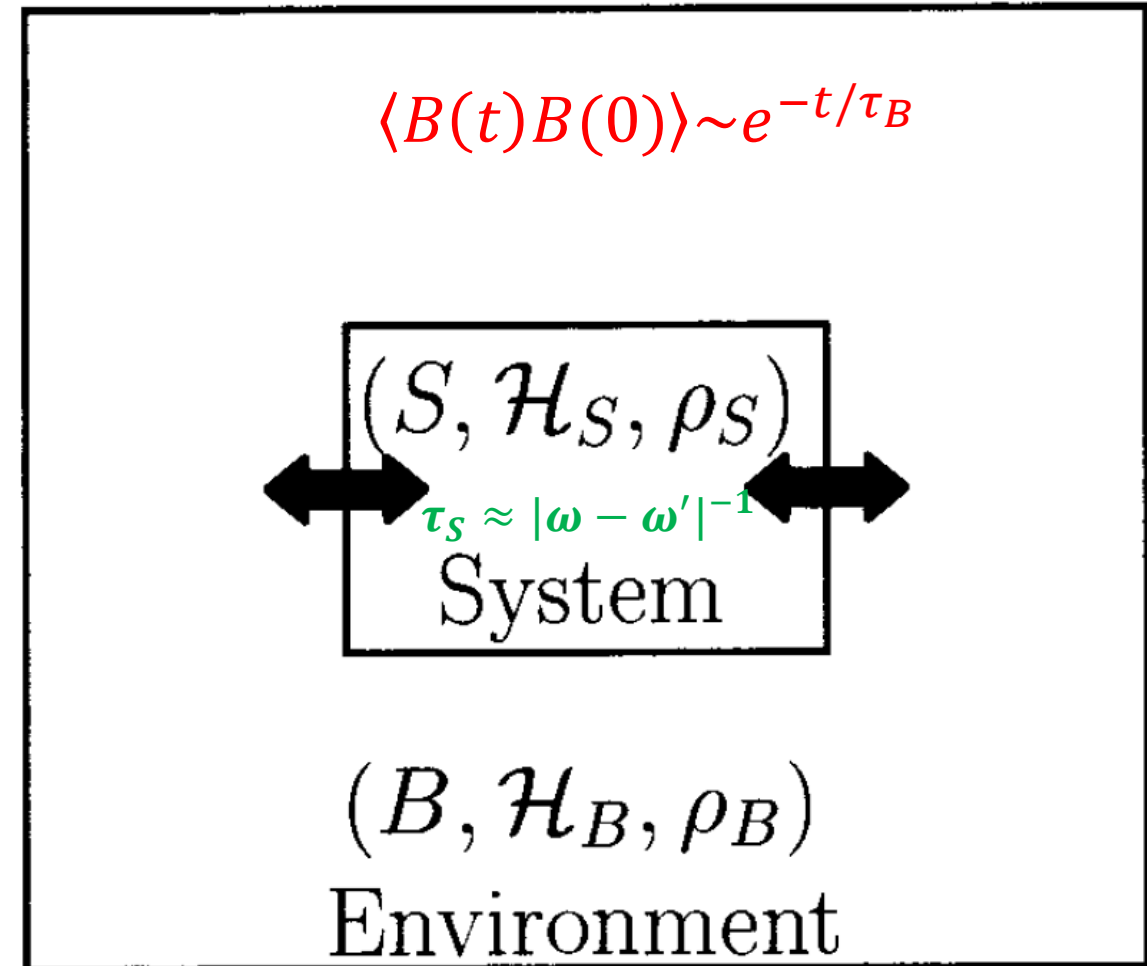
3	Quantum master equations	109
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Quantum master equations: Microscopic derivation



$\tau_S, \tau_R \gg \tau_B$ The bath dynamics is fast – faster than any other timescales of the system

$\tau_R \gg \tau_S$ The relaxation dynamics is slow compared to the (pure) dynamics of the system



The Spin boson model

Model:

$$H_s = \epsilon_0 \sigma_z + \Delta \sigma_x$$

$$H_b = \sum_j \frac{1}{2} (p_j^2 + \omega_j^2 q_j^2)$$

$$H_{sb} = \sigma_z \sum_j g_j q_j$$

$$J(\omega) = \frac{\pi}{2} \sum_j \frac{g_j^2}{\omega_j} \delta(\omega - \omega_j)$$



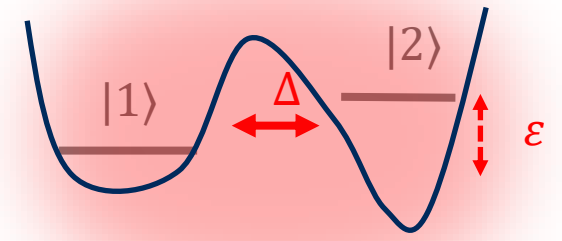
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TORONTO

Dynamics:

Initial condition $\rho_b |1\rangle \langle 1|$

$$\langle \sigma_z(t) \rangle = \text{Tr} \left\{ \rho_0 e^{iHt} \hat{\sigma}_z e^{-iHt} \right\}$$

$$H_I = \sum_{\alpha} \underbrace{A_{\alpha}}_{\text{System}} \otimes \underbrace{B_{\alpha}}_{\text{bath}}$$



Quantum master Equations: Microscopic Derivation

$$(S + B, \mathcal{H}_S \otimes \mathcal{H}_B, \rho)$$

Hamiltonian

$$H = H_S + H_B + H_I \quad (1)$$

Quantum Liouville Equation

$$\frac{d}{dt}\rho(t) = -i[H_I(t), \rho(t)] \quad (2)$$

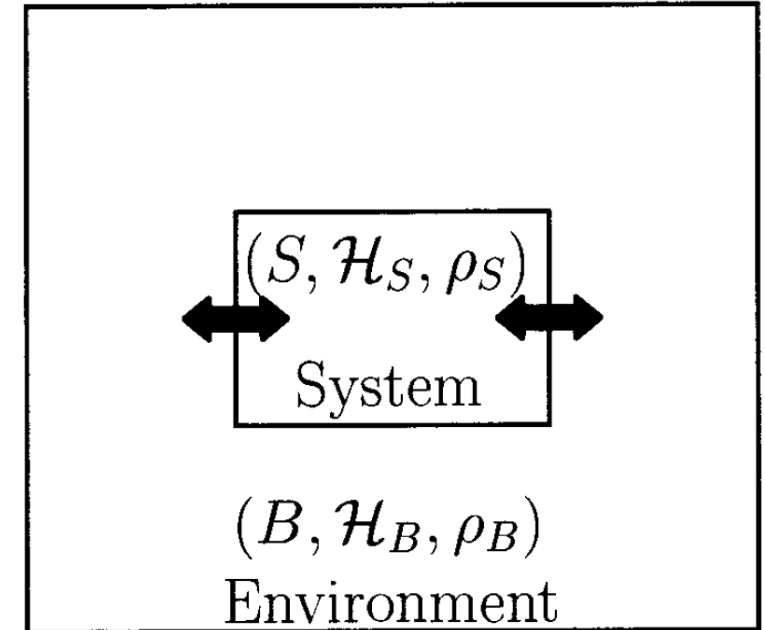
Formal solution:

$$\rho(t) = \rho(0) - i \int_0^t ds [H_I(s), \rho(s)] \quad (3)$$

Plug in Eq. (2) and trace over bath:

$$\frac{d}{dt}\rho_S(t) = - \int_0^t ds \text{tr}_B [H_I(t), [H_I(s), \rho(s)]] \quad (4)$$

Assumption #1 $\text{tr}_B [H_I(t), \rho(0)] = 0$



$$\frac{d}{dt}\rho_S(t) = - \int_0^t ds \text{tr}_B [H_I(t), [H_I(s), \rho_S(s) \otimes \rho_B]] \quad (5)$$

Time-local, Markovian QME:

$$\frac{d}{dt}\rho_S(t) = - \int_0^\infty ds \text{tr}_B [H_I(t), [H_I(t-s), \rho_S(t) \otimes \rho_B]] \quad (6)$$

Assumption #2:

Born approximation

$$\rho(t) \approx \rho_S(t) \otimes \rho_B$$

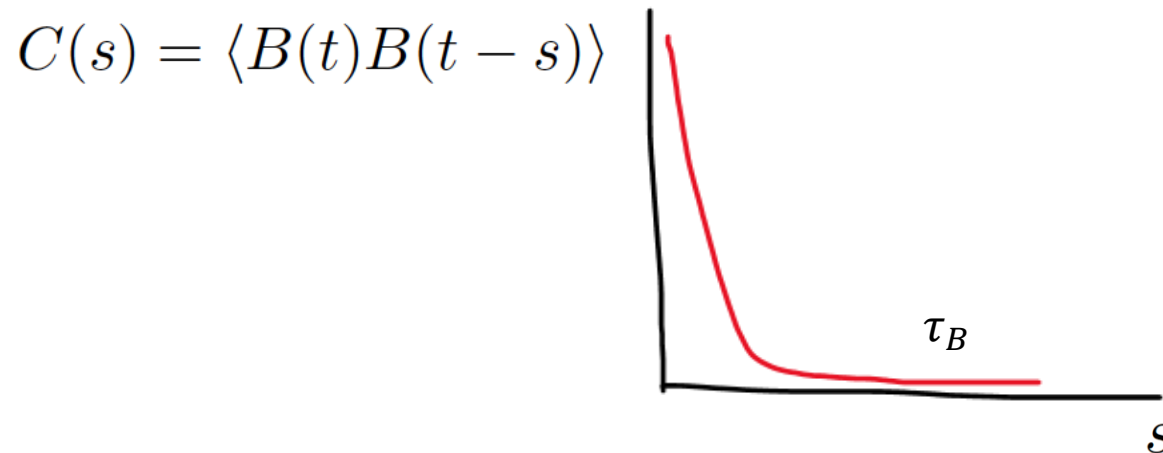
Assumption #3

Markov approximation

Assumption #4

$$H_I = \sum_{\alpha} A_{\alpha} \otimes B_{\alpha}$$

System bath



$$\frac{d}{dt}\rho_S(t) = - \int_0^t ds \text{tr}_B [H_I(t), [H_I(s), \rho_S(s) \otimes \rho_B]] \quad (5)$$

Assumption #2:

Born approximation

$$\rho(t) \approx \rho_S(t) \otimes \rho_B$$

Time-local, Markovian QME:

$$\frac{d}{dt}\rho_S(t) = - \int_0^\infty ds \text{tr}_B [H_I(t), [H_I(t-s), \rho_S(t) \otimes \rho_B]] \quad (6)$$

Assumption #3

Markov approximation

Define “Jump operators”: Summation over Bohr frequencies

$$H_I = \sum_{\alpha} A_{\alpha} \otimes B_{\alpha}$$

Assumption #4

$$A_{\alpha}(\omega) \equiv \sum_{\varepsilon' - \varepsilon = \omega} \Pi(\varepsilon) A_{\alpha} \Pi(\varepsilon')$$

From Assumption 1:

$$\langle B_{\alpha}(t) \rangle \equiv \text{tr} \{ B_{\alpha}(t) \rho_B \} = 0$$

Interaction representation

$$e^{iH_S t} A_{\alpha}(\omega) e^{-iH_S t} = e^{-i\omega t} A_{\alpha}(\omega)$$

Markovian Redfield QME

Time local,
Markovian, but not
GKLS form!

$$\begin{aligned}\frac{d}{dt}\rho_S(t) &= \int_0^\infty ds \operatorname{tr}_B \{ H_I(t-s)\rho_S(t)\rho_B H_I(t) - H_I(t)H_I(t-s)\rho_S(t)\rho_B \} + \text{h.c.} \\ &= \sum_{\omega, \omega'} \sum_{\alpha, \beta} e^{i(\omega' - \omega)t} \Gamma_{\alpha\beta}(\omega) (A_\beta(\omega)\rho_S(t)A_\alpha^\dagger(\omega') - A_\alpha^\dagger(\omega')A_\beta(\omega)\rho_S(t)) \\ &\quad + \text{h.c.}\end{aligned}\tag{3.132}$$

Rates: $\Gamma_{\alpha\beta}(\omega) \equiv \int_0^\infty ds e^{i\omega s} \langle B_\alpha^\dagger(t) B_\beta(t-s) \rangle$

Assumption 5: Stationary bath

$$\langle B_\alpha^\dagger(t) B_\beta(t-s) \rangle = \langle B_\alpha^\dagger(s) B_\beta(0) \rangle$$

Bath-bath Correlation functions:

$$\langle B_\alpha^\dagger(t) B_\beta(t-s) \rangle \equiv \operatorname{tr}_B \{ B_\alpha^\dagger(t) B_\beta(t-s) \rho_B \}$$

Lindblad QME

$$\tau_S, \tau_R \gg \tau_B$$

The bath dynamics is fast – faster than any other timescales of the system

$$\langle B(t)B(0) \rangle \sim e^{-t/\tau_B}$$

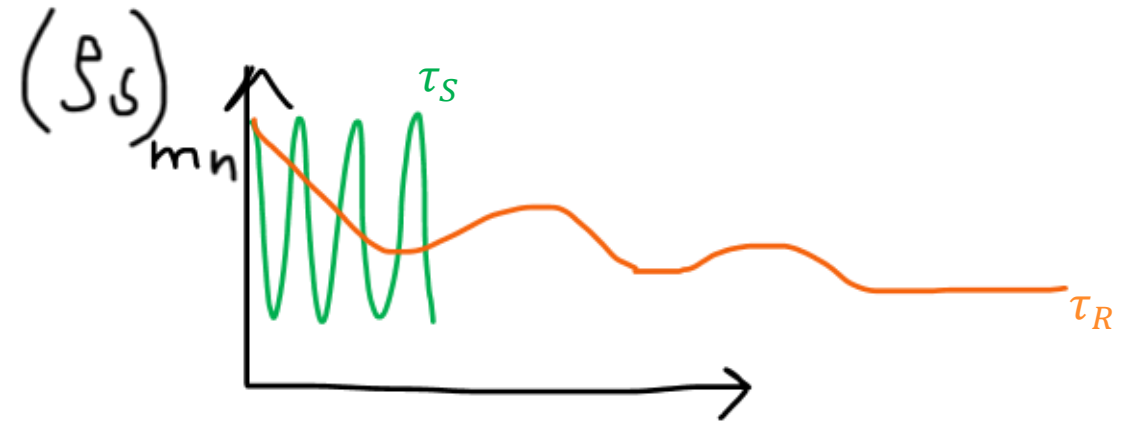
$$\tau_R \gg \tau_S$$

The relaxation dynamics is slow compared to the (pure) dynamics of the system

$$\tau_S \approx |\omega - \omega'|^{-1}$$

Lindblad form

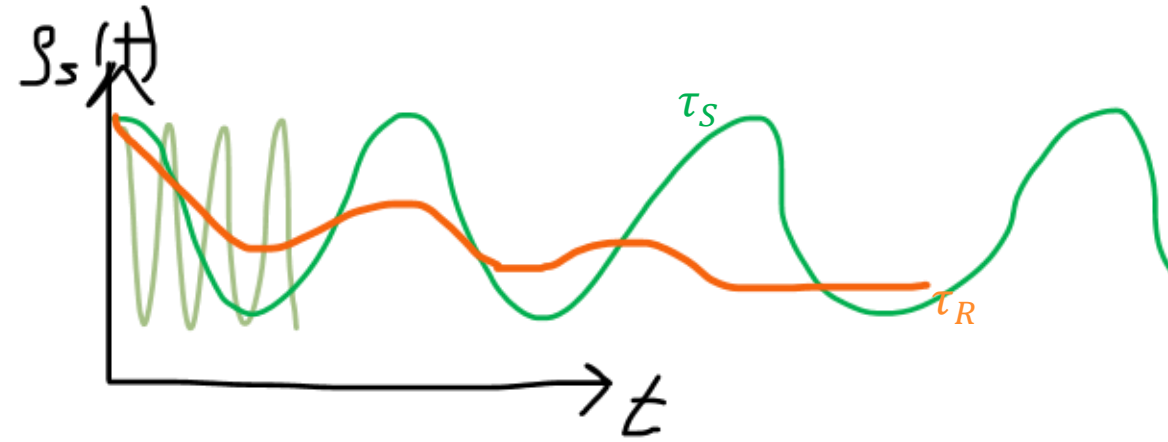
Assumption #6: Rotating wave approximation



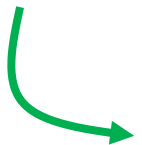
$$\frac{d}{dt}\rho_S(t) = \sum_{\omega} \sum_{\alpha, \beta} \Gamma_{\alpha\beta}(\omega) (A_{\beta}(\omega)\rho_S(t)A_{\alpha}^{\dagger}(\omega) - A_{\alpha}^{\dagger}(\omega)A_{\beta}(\omega)\rho_S(t)) + \text{h.c.}$$

Unified QME

Maintains coupling between coherences and population in the energy basis – better accuracy than the fully secular QME for systems with near degeneracy



$$\frac{d}{dt}\rho_S(t) = -i[H_{LS}, \rho_S(t)] + \sum_{\omega, \omega'} \sum_{\alpha, \beta} \gamma_{\alpha\beta}(\omega, \omega') e^{i(\omega' - \omega)t} \left(A_{\beta\omega} \rho_S(t) A_{\alpha\omega'}^\dagger - \frac{1}{2} \{ A_{\alpha\omega'}^\dagger, A_{\beta\omega}, \rho_S(t) \} \right)$$



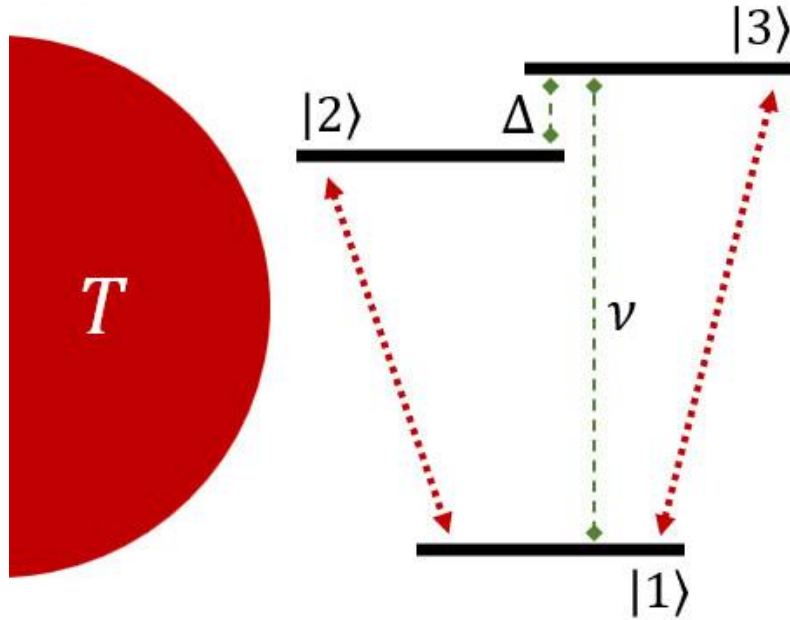
$$\gamma_{\alpha\beta}(\omega, \omega') \approx \gamma_{\alpha\beta}(\bar{\omega}, \bar{\omega}) \equiv \gamma_{\alpha\beta}(\bar{\omega})$$

Assumption #6:
partial Rotating wave approximation

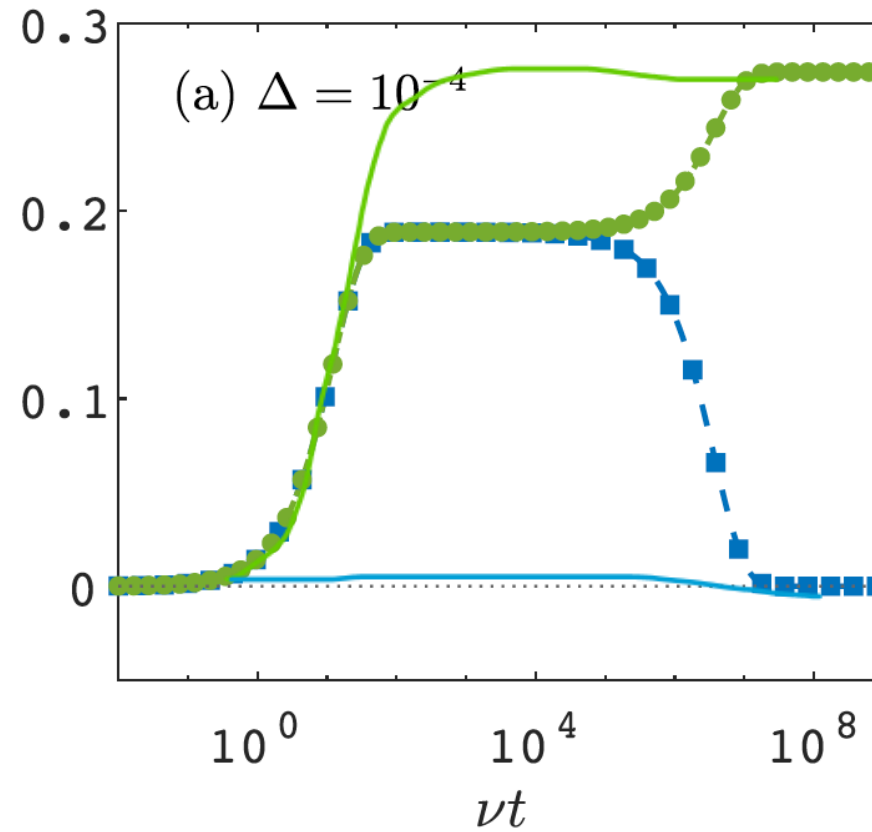
$$\frac{d}{dt}\rho_S(t) = -i[H_{LS}, \rho_S(t)] + \sum_{\bar{\omega}} \sum_{\omega, \omega' \in \mathcal{F}_{\bar{\omega}}} \sum_{\alpha, \beta} e^{i(\omega' - \omega)t} \gamma_{\alpha\beta}(\bar{\omega}) \left(A_{\beta\omega} \rho_S(t) A_{\alpha\omega'}^\dagger - \frac{1}{2} \{ A_{\alpha\omega'}^\dagger, A_{\beta\omega}, \rho_S(t) \} \right)$$

Hyperacceleration of quantum thermalization dynamics by bypassing long-lived coherences: An analytical treatment

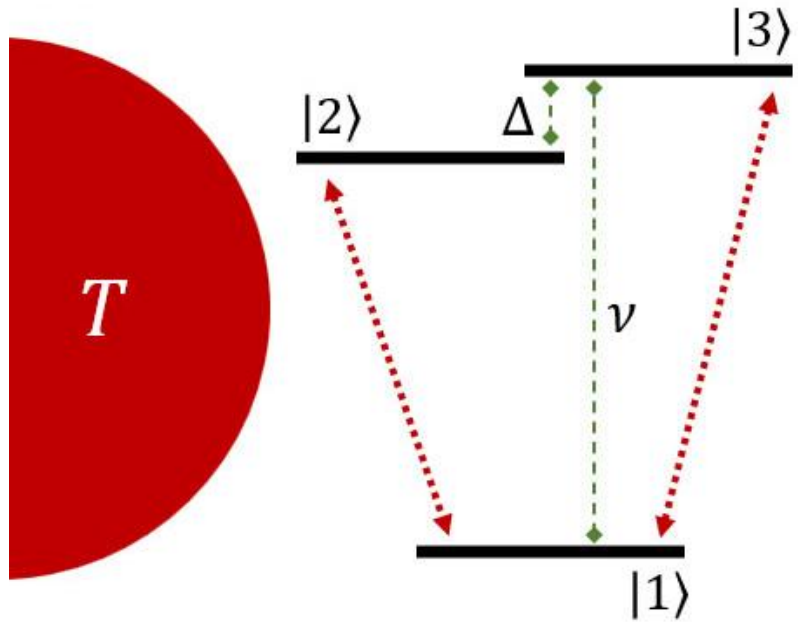
Redifeld QME $\rightarrow$ Unified QME $\rightarrow$ Lindblad QME



- Redfield $\sigma_{32}^R(t)$
- Redfield $P(t)$
- Unified $\sigma_{32}^R(t)$
- Unified $P(t)$

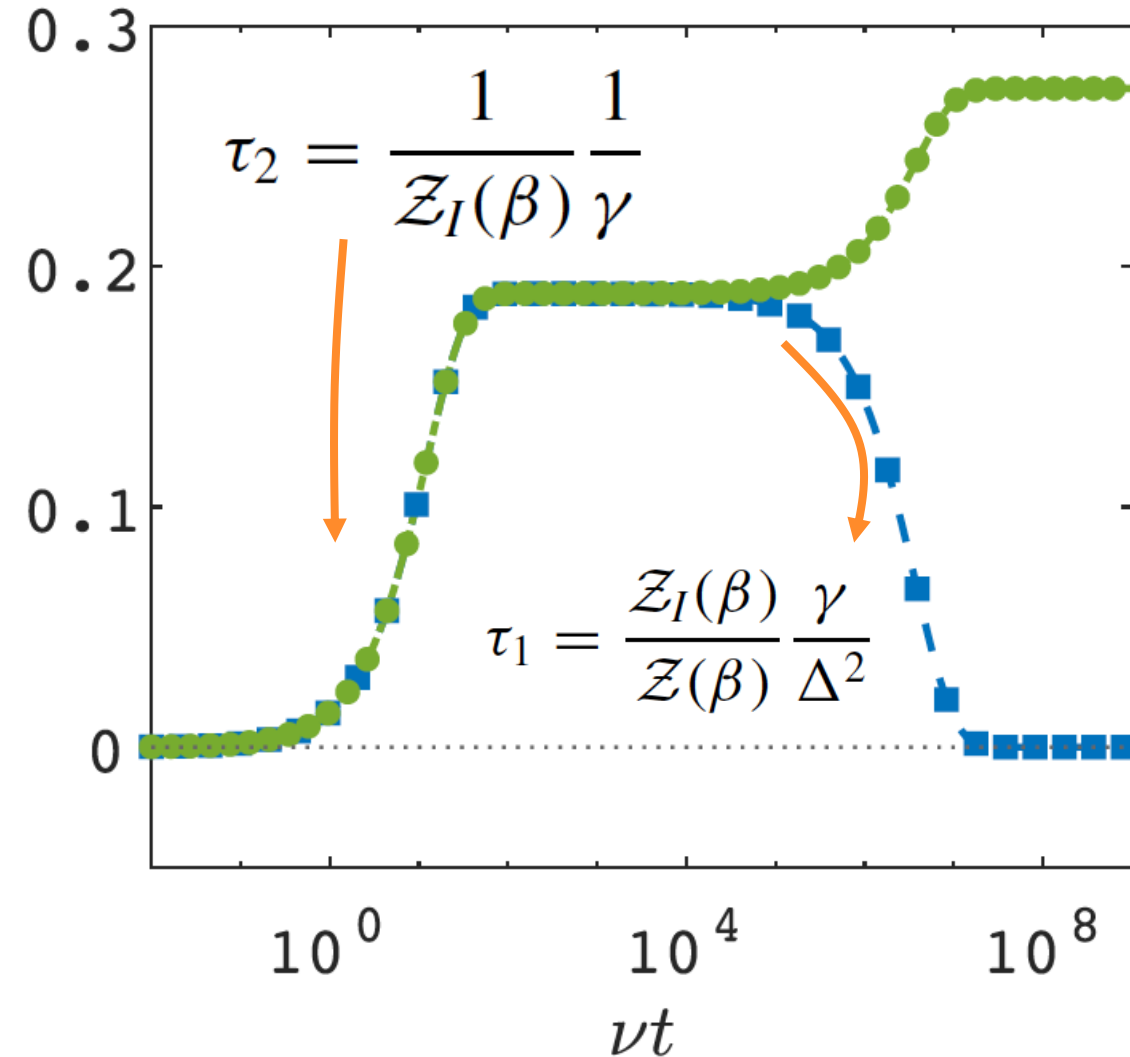


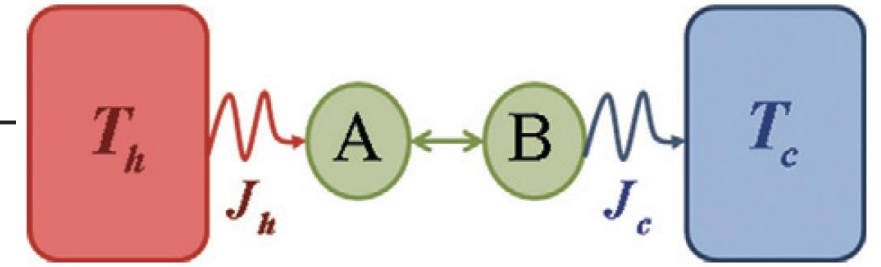
Hyperacceleration of quantum thermalization dynamics by bypassing long-lived coherences: An analytical treatment



$$Z_I(\beta) = 1 + e^{-\beta\nu}$$

$$Z(\beta) = 1 + 2e^{-\beta\nu}$$





The local approach to quantum transport may violate the second law of thermodynamics

AMIKAM LEVY and RONNIE KOSLOFF

Institute of Chemistry, The Hebrew University of Jerusalem - Jerusalem 91904, Israel

$$\mathcal{J}_h = (e^{\beta_c \omega_c} - e^{\beta_h \omega_h}) \mathcal{F} \quad \text{Local}$$

$$\begin{aligned} \mathcal{J}_h = & \frac{(e^{\beta_c \omega_-} - e^{\beta_h \omega_-}) \gamma_c^- \gamma_h^- \omega_-}{\sin^{-2}(\theta) e^{\beta_h \omega_-} (-1 + e^{\beta_c \omega_-}) \gamma_c^- + e^{\beta_c \omega_+} (-1 + e^{\beta_h \omega_-}) \cos^{-2}(\theta) \gamma_h^-} \\ & + \frac{(e^{\beta_c \omega_+} - e^{\beta_h \omega_+}) \gamma_c^+ \gamma_h^+ \omega_+}{e^{\beta_h \omega_+} (-1 + e^{\beta_c \omega_+}) \cos^{-2}(\theta) \gamma_c^+ + \sin^{-2}(\theta) e^{\beta_c \omega_+} (-1 + e^{\beta_h \omega_+}) \gamma_h^+} \end{aligned}$$

global

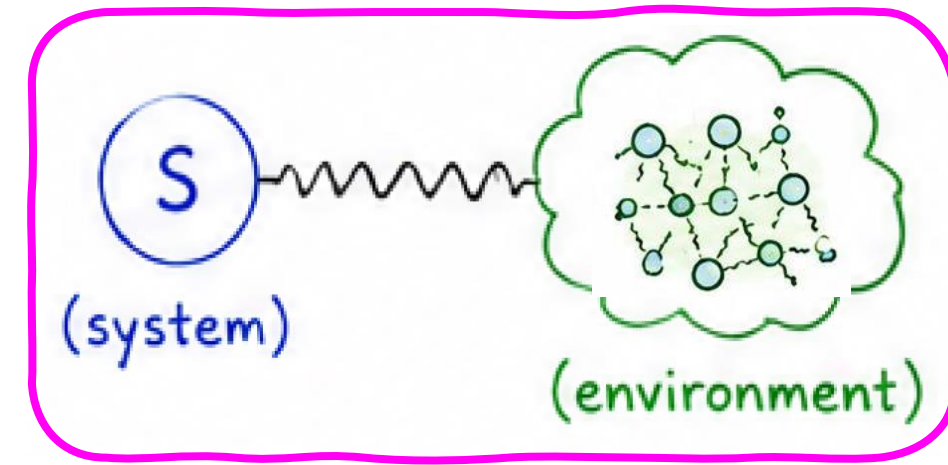
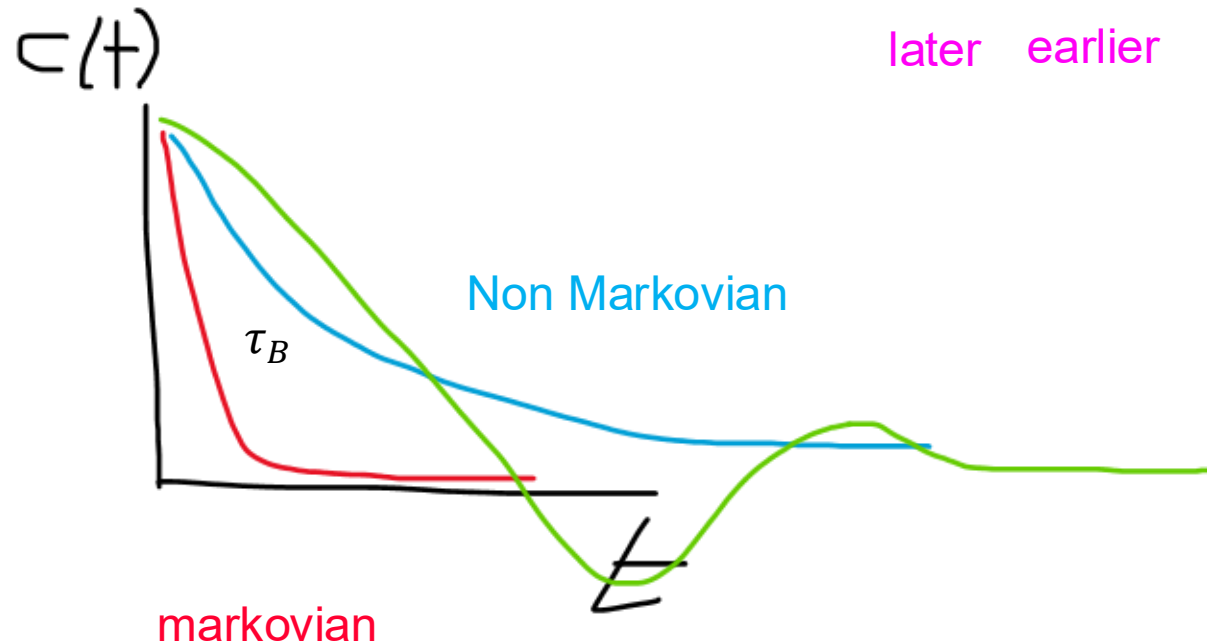
Bath correlation function

Rates: $\Gamma_{\alpha\beta}(\omega) \equiv \int_0^\infty ds e^{i\omega s} \langle B_\alpha^\dagger(t) B_\beta(t-s) \rangle$

$$C(t) = \langle B^\dagger(t) B(0) \rangle_\beta$$

later earlier

Thermal average



$$H = H_S + H_B + H_{SB}$$

$$H_{SB} = S \otimes B$$

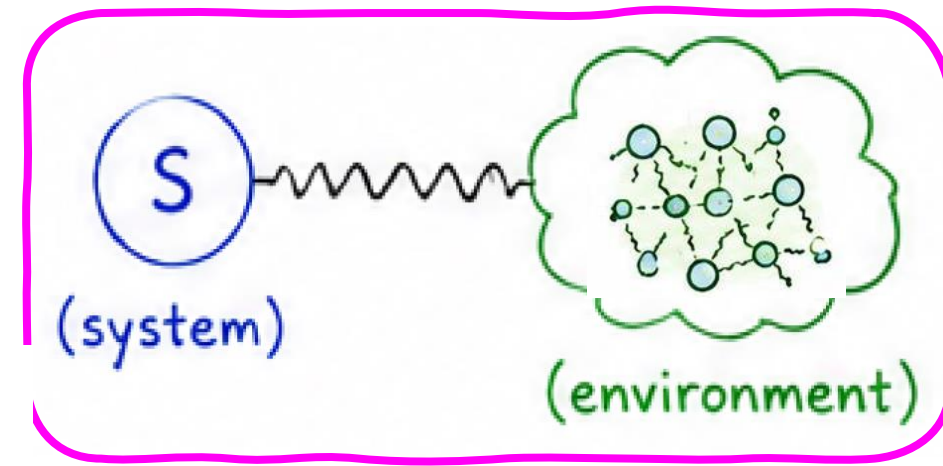
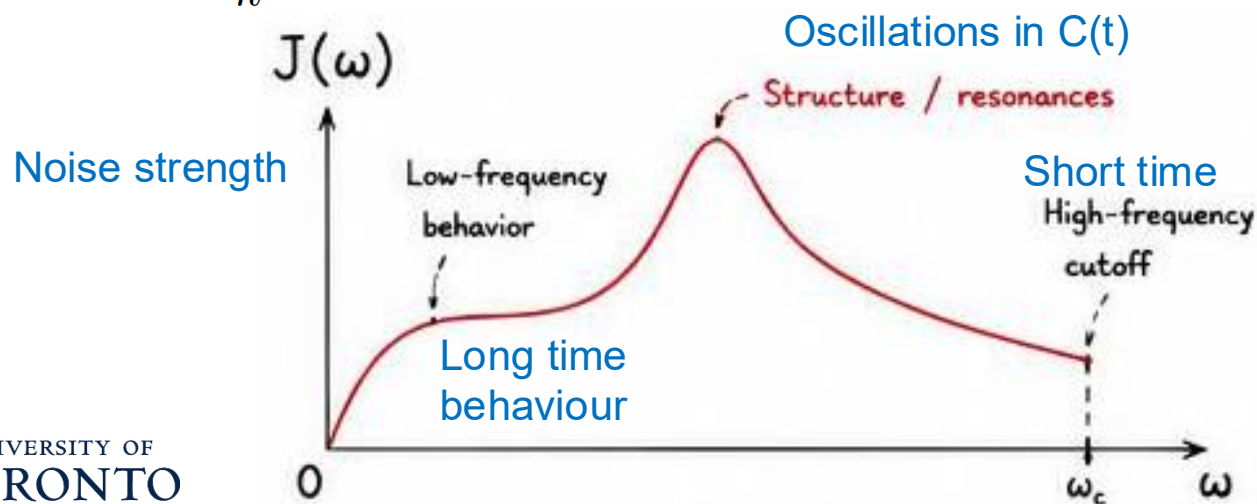
- Memory time τ_B . Long memory time $\rightarrow$ strong memory = non Markovian dynamics
- Characteristic frequency
- Noise strength

Bath correlation function $C(t) = \langle B^\dagger(t)B(0) \rangle_\beta$

$$C(t) = \sum_k g_k^2 [(n_k + 1)e^{-i\omega_k t} + n_k e^{i\omega_k t}]$$

$$C(t) = \frac{1}{\pi} \int_0^\infty d\omega J(\omega) \left[\coth\left(\frac{\beta\omega}{2}\right) \cos(\omega t) - i \sin(\omega t) \right]$$

$$J(\omega) = \pi \sum_k g_k^2 \delta(\omega - \omega_k)$$



$$H = H_S + H_B + H_{SB}$$

$$H_B = \sum_k \omega_k b_k^\dagger b_k$$

$$B = \sum_k g_k (b_k^\dagger + b_k)$$

Outline: Today's talk #2

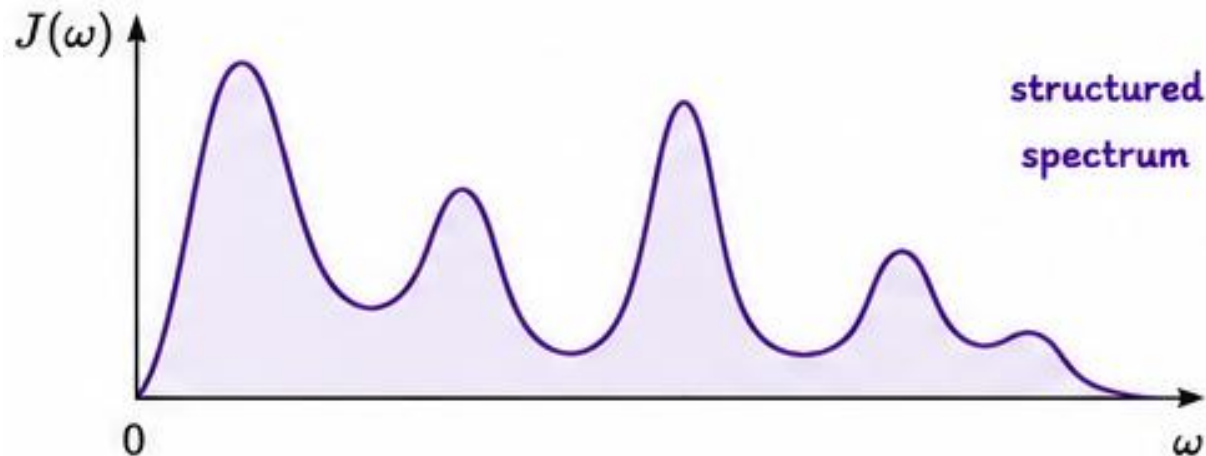
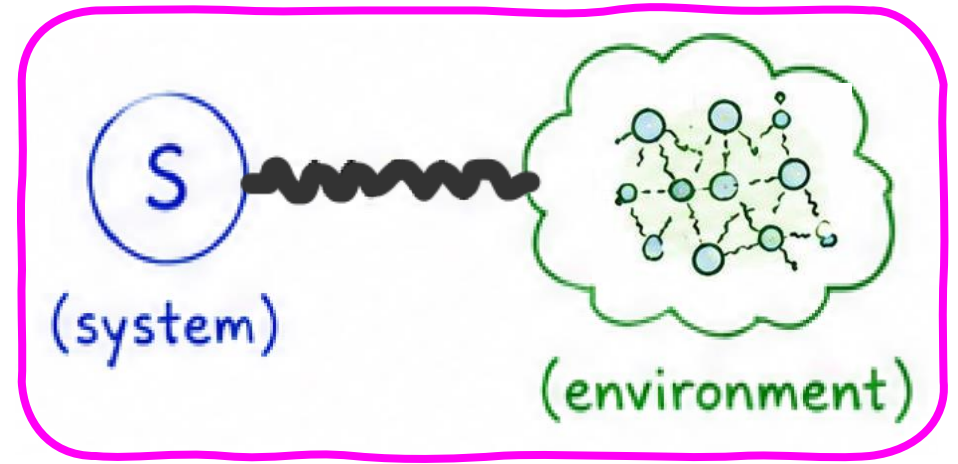
○ Review

- Weak coupling, Markovian approximation
- Quantum master equation

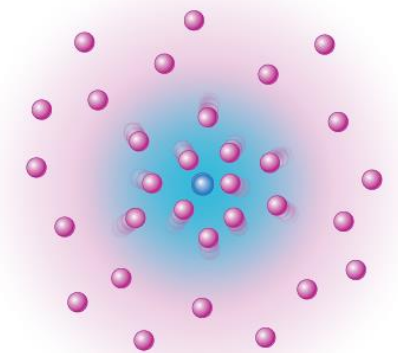
○ Memory effects: Measures, Methods

○ Strong coupling: Polaron mapping

○ Other methods: Repeated interaction, Path integral, HEOM,...



A diagram illustrating the polaron mapping. It shows a 4x4 grid of purple dots representing lattice sites. A blue dot is located in the center of the grid, representing the polaron. The blue dot is surrounded by a blue cloud, indicating its interaction with the lattice.



F. Chevy, Physics 9, 86 (2016)

The Nakajima–Zwanzig equation: Exact formalism

$$\frac{d\rho_S(t)}{dt} = \int_0^t \mathcal{K}(t - \tau) \rho_S(\tau) d\tau + I(t)$$

Memory Kernel
(non Markovian)

Initial system-bath
correlations

Markovian Limit
(memoryless)

$$\frac{d\rho_S(t)}{dt} = \mathcal{L}[\rho_S(t)]$$

The Nakajima–Zwanzig equation: Exact formalism

$$\frac{d}{dt}\hat{\rho}(t) = -\frac{i}{\hbar} [\hat{H}, \hat{\rho}(t)]$$

$$\hat{P} + \hat{Q} = 1$$

$$\equiv -i\mathcal{L}\hat{\rho}(t)$$

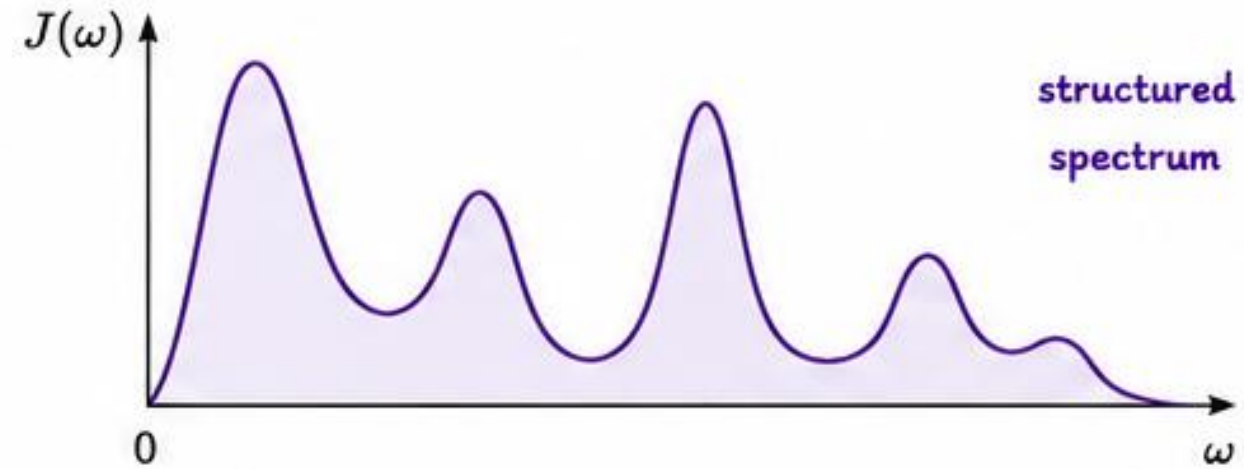
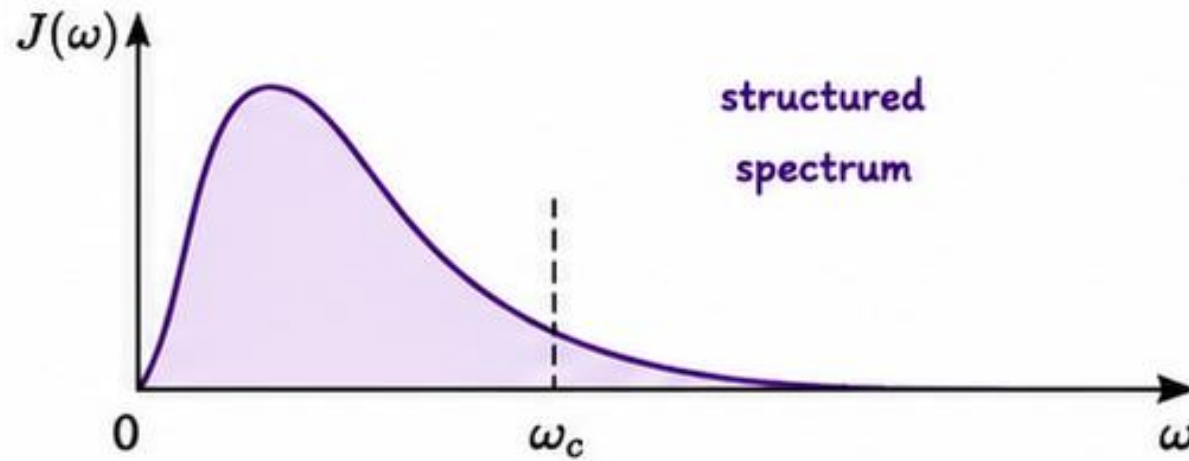
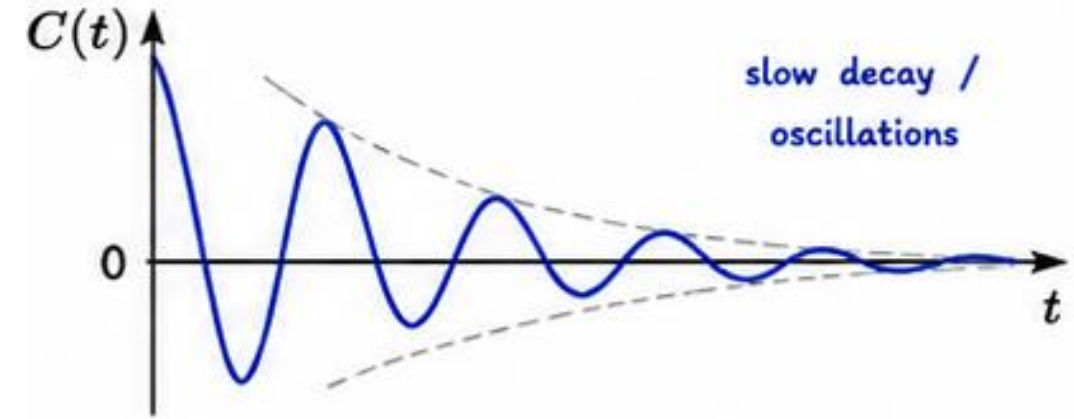
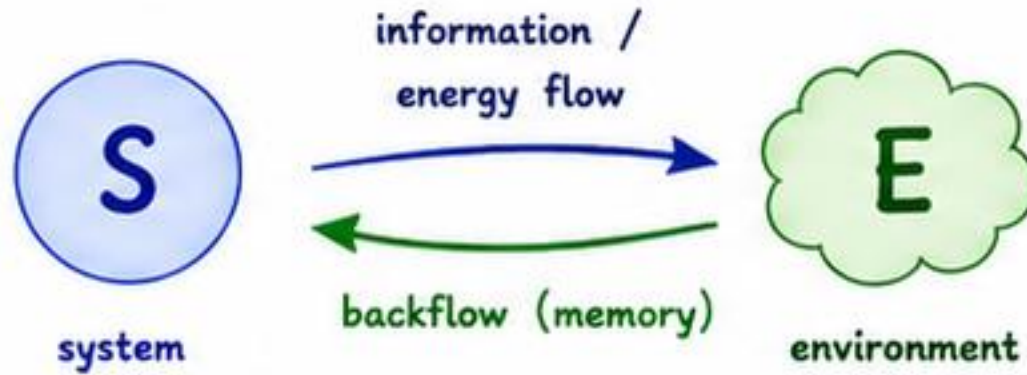
$$\hat{P}\hat{\rho} \equiv \hat{\rho}_b \text{Tr}_B \hat{\rho} = \hat{\rho}_b \hat{\rho}_S$$

$$\frac{d}{dt}\hat{P}\hat{\rho} = -i\hat{P}\mathcal{L}\hat{\rho} = -i\hat{P}\mathcal{L}\hat{P}\hat{\rho} - i\hat{P}\mathcal{L}\hat{Q}\hat{\rho}$$

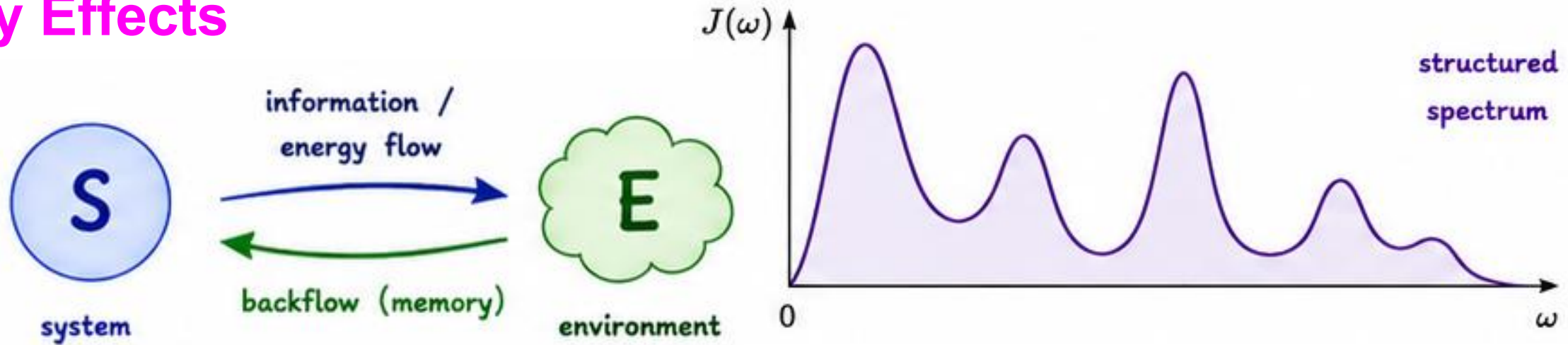
$$\frac{d}{dt}\hat{Q}\hat{\rho} = -i\hat{Q}\mathcal{L}\hat{\rho} = -i\hat{Q}\mathcal{L}\hat{P}\hat{\rho} - i\hat{Q}\mathcal{L}\hat{Q}\hat{\rho}$$

$$\frac{d}{dt}\hat{P}\hat{\rho}(t) = -i\hat{P}\mathcal{L}\hat{P}\hat{\rho}(t) - \underbrace{\int_{t_0}^t d\tau \hat{P}\mathcal{L}e^{-i\hat{Q}\mathcal{L}(t-\tau)} \hat{Q}\mathcal{L}\hat{P}\hat{\rho}(\tau)}_{\text{Memory Kernel (non Markovian)}} - \underbrace{i\hat{P}\mathcal{L}e^{-i\hat{Q}\mathcal{L}(t-t_0)} \hat{Q}\hat{\rho}(t_0)}_{\text{Initial correlations}}$$

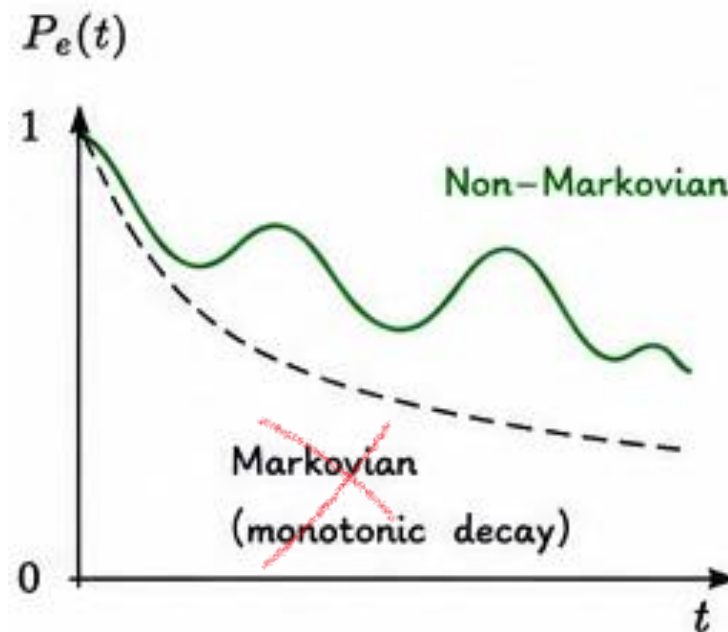
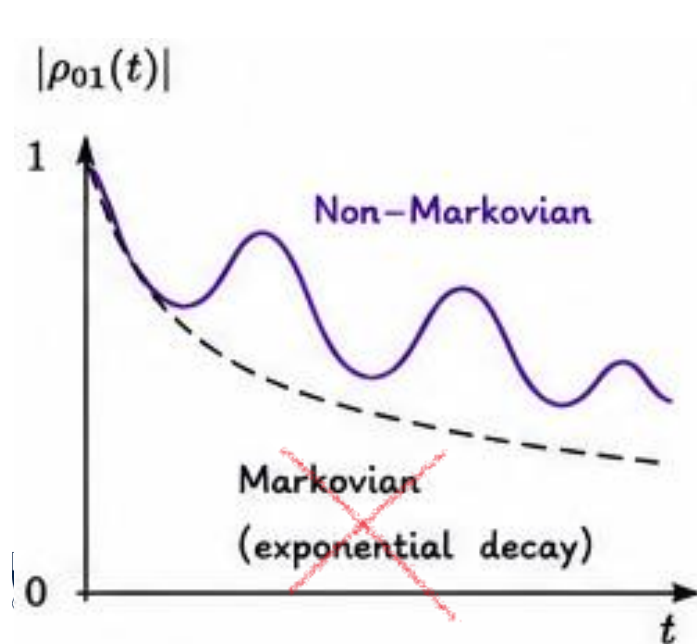
Memory Effects



Memory Effects



$$\frac{d\rho_S(t)}{dt} = \int_0^t \mathcal{K}(t - \tau) \rho_S(\tau) d\tau + I(t)$$



Dynamics of non-Markovian open quantum systems

Inés de Vega

*Department of Physics and Arnold Sommerfeld Center for Theoretical Physics,
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Daniel Alonso

*Instituto de Estudios Avanzados (IUdEA) and Departamento de Física,
Universidad de La Laguna, La Laguna 38204, Tenerife, Spain*

$$\frac{d}{dt}\rho(t) = \mathcal{L} \int_0^t k(\tau) e^{\mathcal{L}\tau} \rho(t - \tau) d\tau$$

$\mathcal{L}$ is a (Markovian) Lindblad generator and $k(\tau)$ is a phenomenological memory kernel.

A. Shabani and D. A. Lidar, Completely positive postmarkovian master equation via a measurement approach, Phys. Rev. A 71, 020101(R) (2005).

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Memory Effects

Probing non-Markovian qubit noise and modeling Post Markovian Master Equation

arXiv:2510.12894

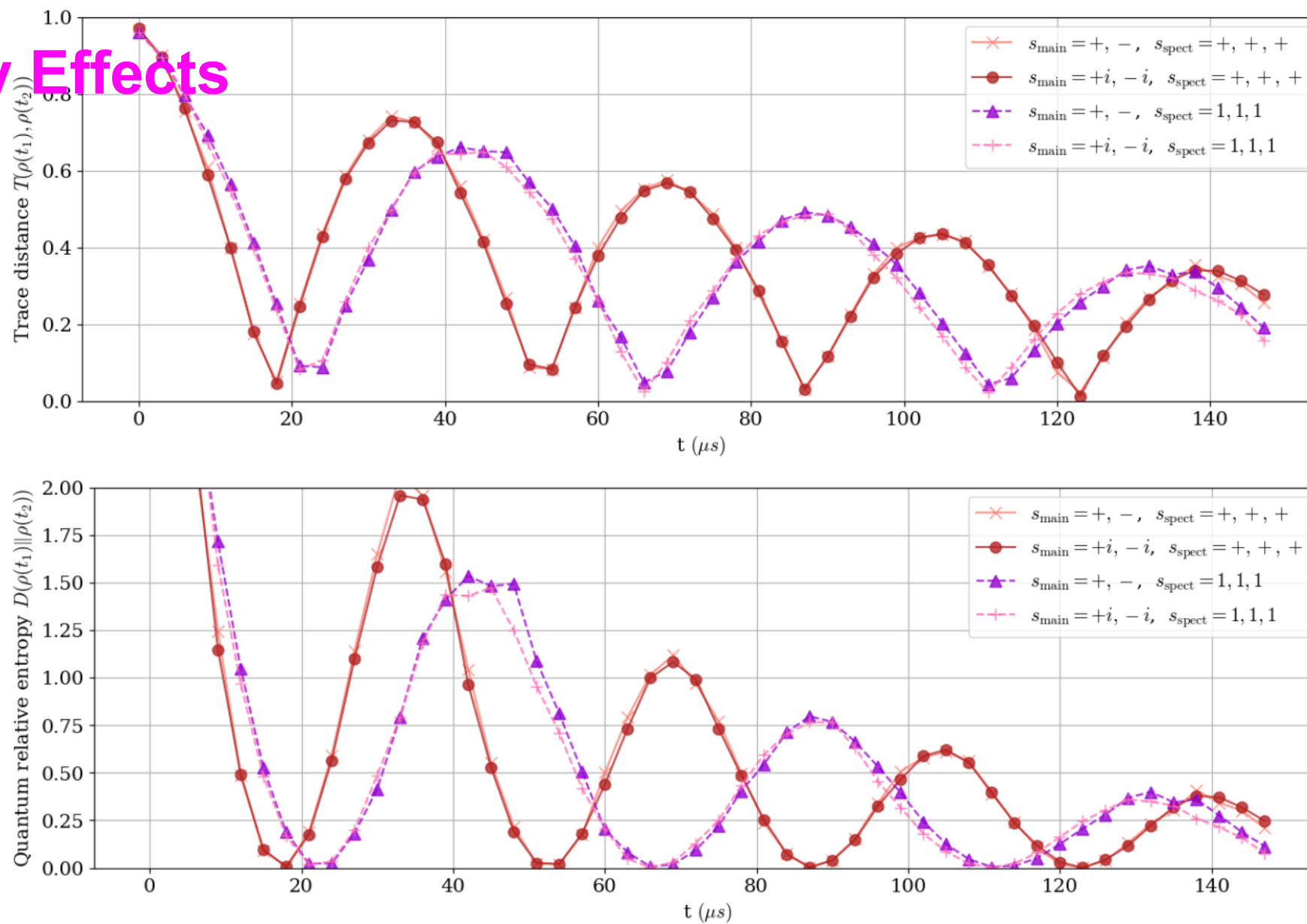
Chun-Tse Li,¹ Jingming Tan,¹ Vasil Gucev,¹ and Daniel Lidar¹

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Understanding the noise characteristics of quantum processors is crucial when achieving fault tolerant quantum computing. However, typical qubit designs are often studied under the Markovian approximation, which does not fully capture realistic dynamics. Factors such as qubit-qubit coupling and extended bath correlation times can introduce significant non-Markovian effects into the noise processes. In this study, we employ the Post-Markovian Master Equation (PMME) formalism to characterize memory effects in the noise dynamics. We further experimentally validate the PMME framework using superconducting qubits on an IBM Quantum device, demonstrating clear non-Markovian behavior during circuit execution. Additionally, we quantify the crosstalk effect using an information-theoretic approach and reveal that crosstalk can dominate the observed non-Markovian effects in current quantum hardware.

Memory Effects

arXiv:2510.12894



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Figure 6: Time evolution of (a) the trace distance $T(\rho_1(t), \rho_2(t))$ and (b) the quantum relative entropy $D(\rho_1(t) \parallel \rho_2(t))$ for the two initial pairs $(|+\rangle, |-\rangle)$ and $(|+i\rangle, |-i\rangle)$. The experiments are conducted on IBM’s 127-qubit “Brussels” processor, the main qubit index is 49 while the spectator qubits index are $\{48, 50, 55\}$. Solid lines correspond to spectator qubits in $|+\rangle^{\otimes 3}$; dashed lines to spectators in $|1\rangle^{\otimes 3}$. The periodic revivals indicate information backflow and thus non-Markovian behavior.

Memory Effects Measure for the Non-Markovianity

Phys. Rev. Lett. 103, 210401 (2009)

► Breuer–Laine–Piilo (BLP) measure

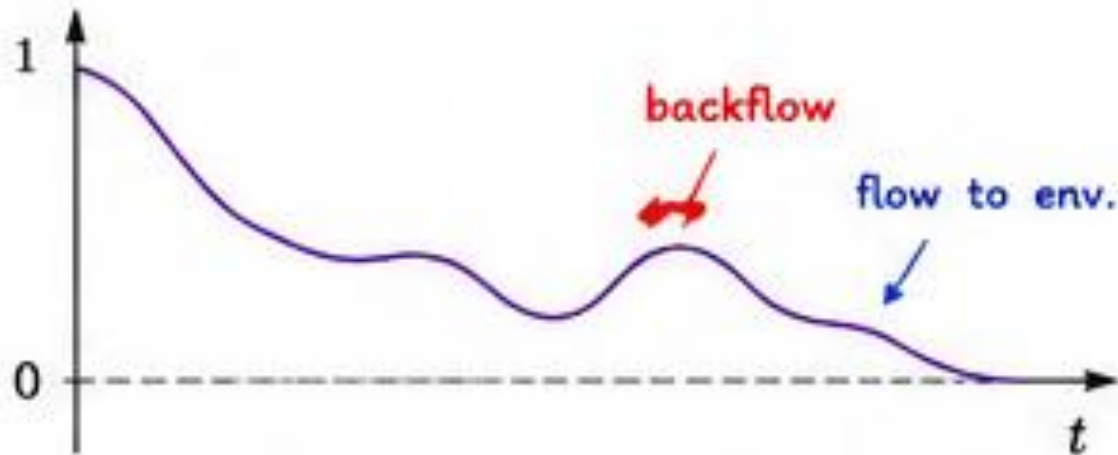
$$D(\rho_1, \rho_2) = \frac{1}{2} \text{tr} |\rho_1 - \rho_2|$$

$$\mathcal{N}_{\text{BLP}} = \max_{\rho_{1,2}} \int_{\sigma(t) > 0} \sigma(t) dt$$

$$\sigma(t) = \frac{d}{dt} D[\rho_1(t), \rho_2(t)]$$

D : trace distance, $\rho_{1,2}(t)$: two initial states

$D[\rho_1(t), \rho_2(t)]$



$\sigma(t) > 0$ indicates
increase of distinguishability
⇒ information backflow.

Memory Effects Measure for the Non-Markovianity

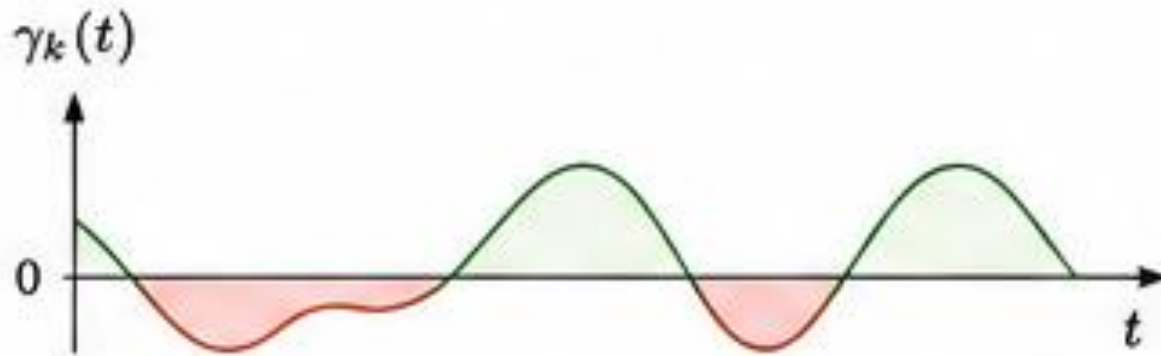
PRL 105, 050403 (2010)

► Rivas–Huelga–Plenio (RHP) measure

$$\mathcal{N}_{\text{RHP}} = \int_{\gamma(t) < 0} (-\gamma(t)) dt$$

$\gamma(t)$: sum of negative decay rates in the time-local master equation

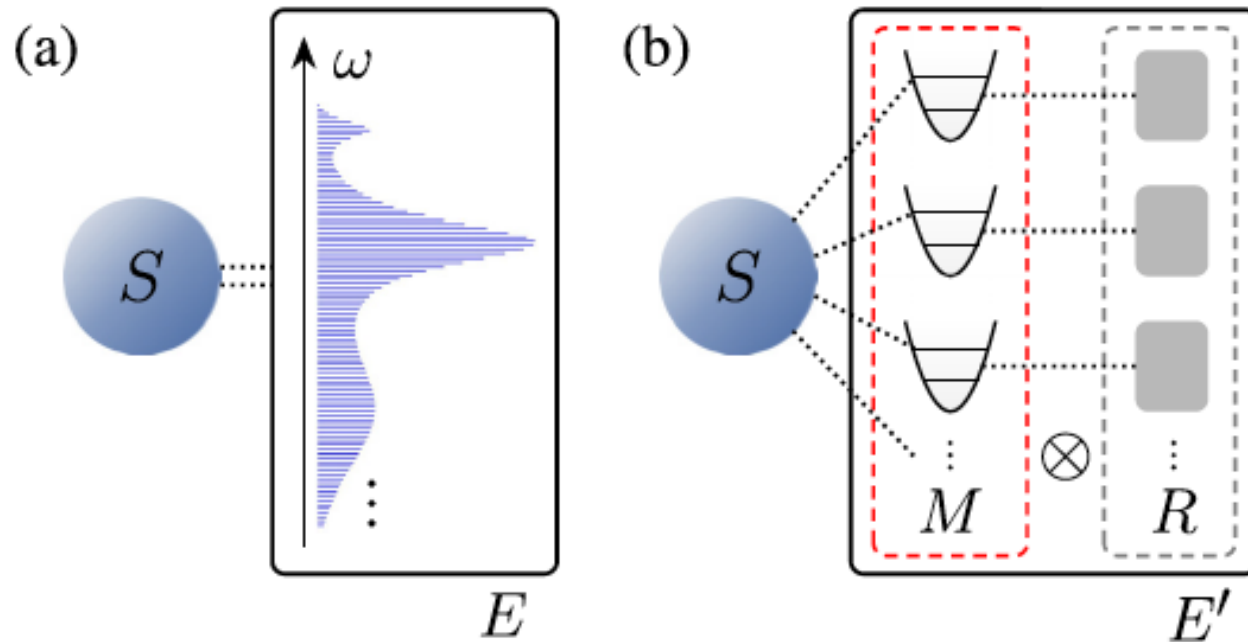
$$\dot{\rho}(t) = \mathcal{L}_t[\rho(t)] = \sum_k \gamma_k(t) \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right)$$



$\gamma_k(t) < 0$ indicates temporary information backflow and loss of CP-divisibility.

Markovian Embedding

- 1) Pseudomodes
- 2) Reaction coordinate mapping
- 3) Chain Mapping

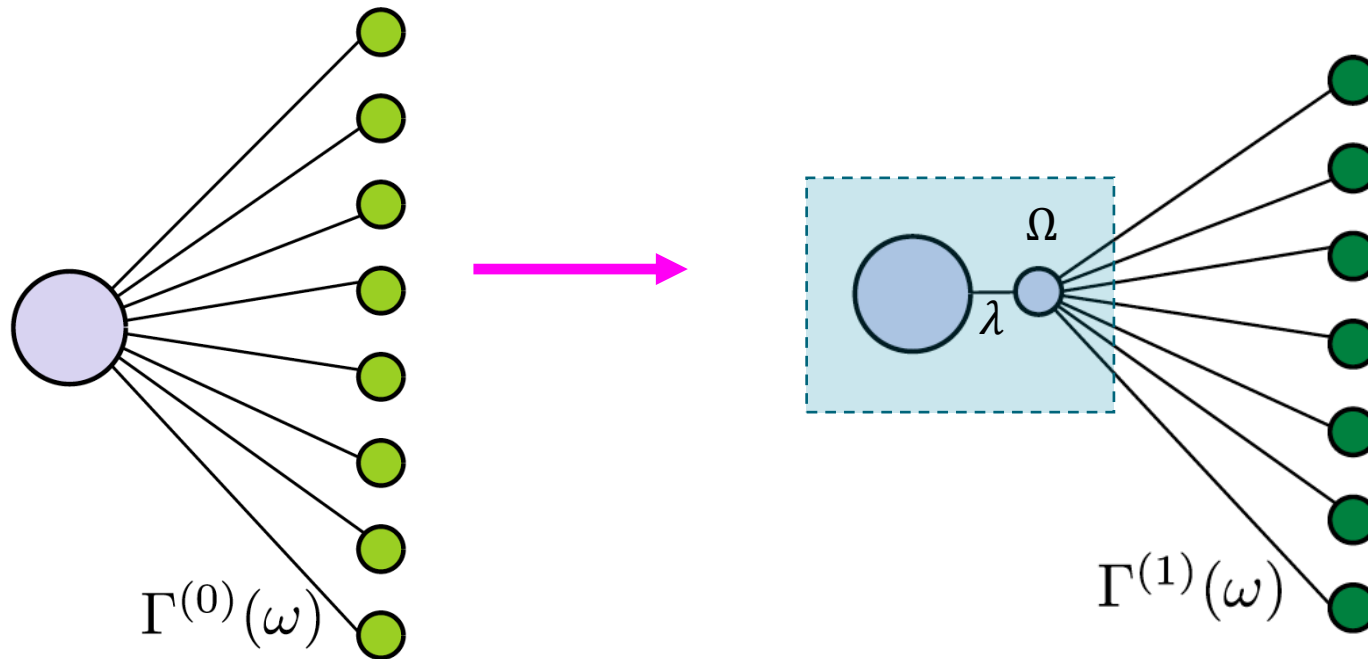


We develop an exact framework for describing the non-Markovian dynamics of an open quantum system interacting with an environment modeled by a generalized spectral density function. The approach relies on mapping the initial system onto an auxiliary configuration, comprising the original open system coupled to a small number of discrete modes, which in turn are each coupled to an independent Markovian reservoir

Phys. Rev. Research **2**, 043058 (2020)

Markovian Embedding

- 1) Pseudomodes
- 2) Reaction coordinate mapping
- 3) Chain Mapping



$$\Gamma^{(0)}(\omega) = 2\pi \sum_k |h_k|^2 \delta(\omega - \omega_k)$$

$$\lambda^2 = \frac{1}{2\pi\Omega} \int_0^\infty \omega \Gamma^{(0)}(\omega) d\omega$$

$$\Omega^2 = \frac{\int_0^\infty \omega^3 \Gamma^{(0)}(\omega) d\omega}{\int_0^\infty \omega \Gamma^{(0)}(\omega) d\omega}$$

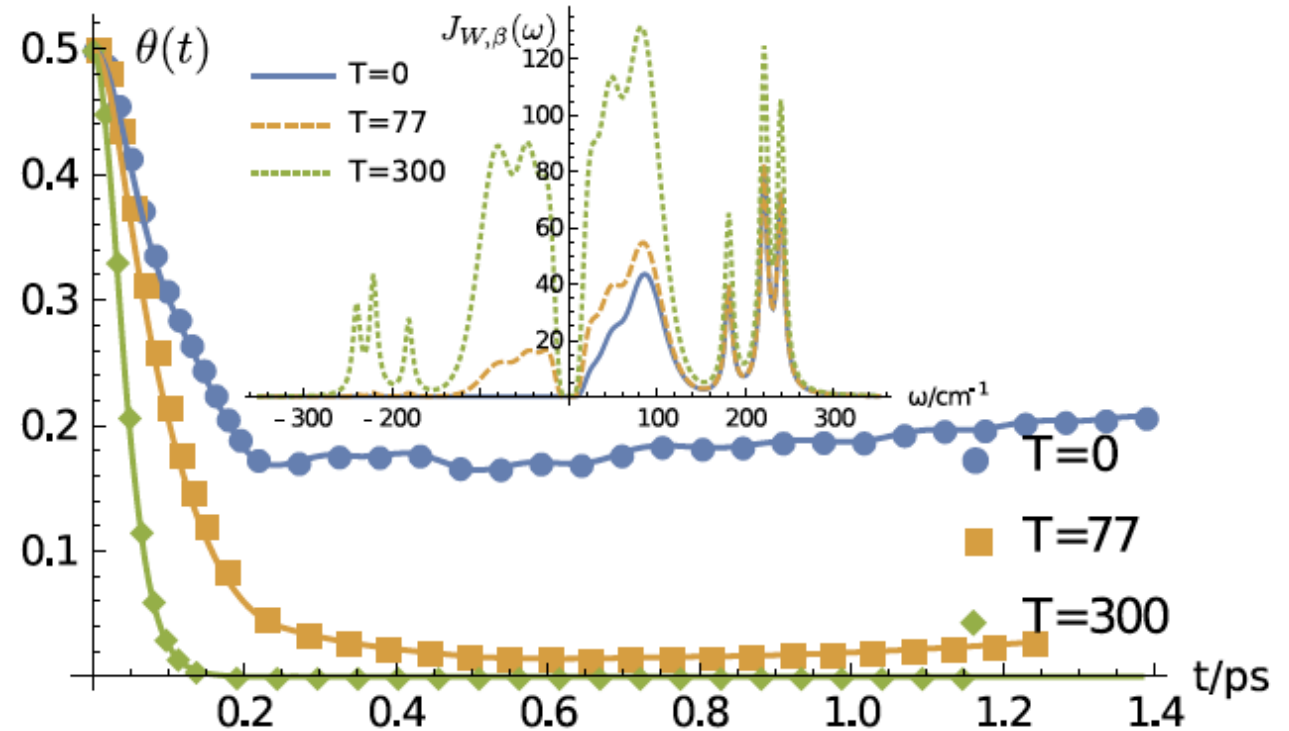
$$\Gamma^{(1)}(\omega) = \frac{4\lambda^2 \Gamma^{(0)}(\omega)}{\left[\frac{1}{\pi} \mathcal{P} \int \frac{\Gamma^{(0)}(\omega')}{\omega' - \omega} d\omega' \right]^2 + [\Gamma^{(0)}(\omega)]^2}$$

Markovian Embedding

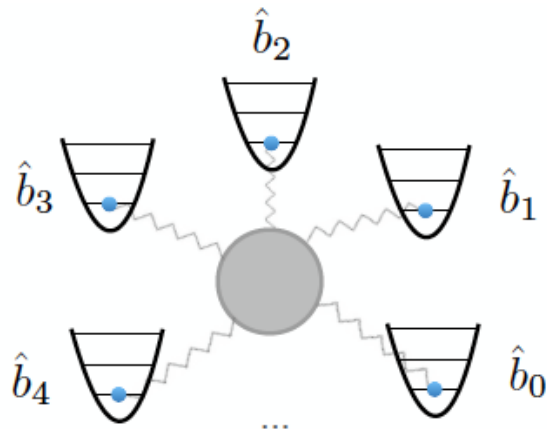
- 1) Pseudomodes
- 2) Reaction coordinate mapping
- 3) Chain Mapping

Exact mapping between system-reservoir quantum models and semi-infinite discrete chains using orthogonal polynomials AW Chin, Á Rivas, SF Huelga, MB Plenio
Journal of Mathematical Physics 51 (9) (2010)

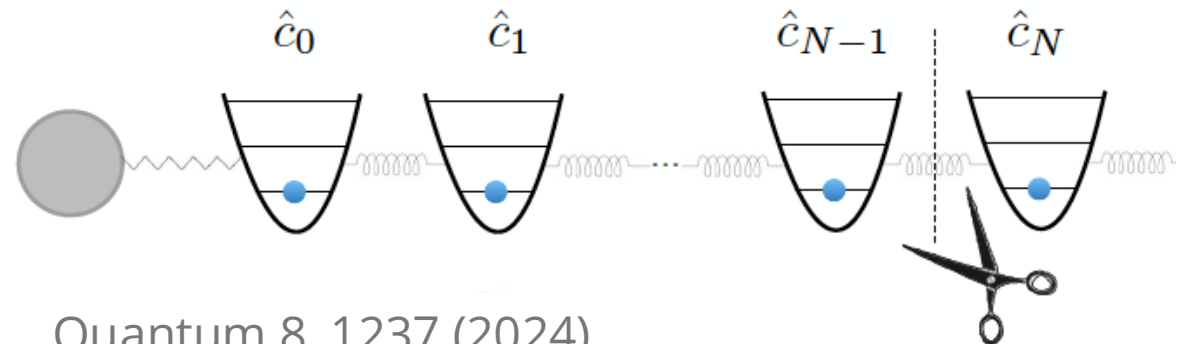
Phys. Rev. Lett. **123**, 090402 (2019)



a)



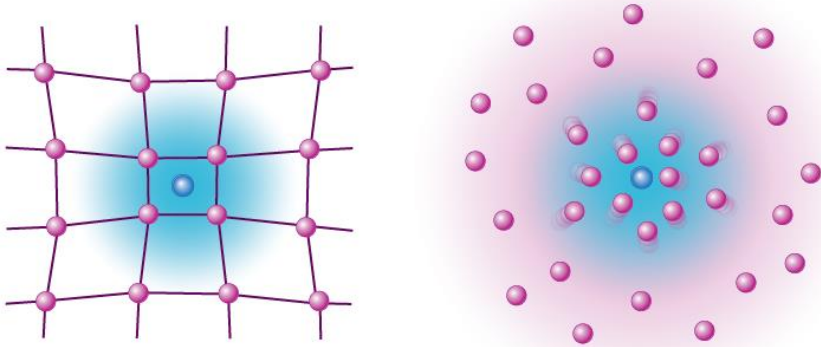
$$\hat{c}_i = \sum_j \alpha_{ij} \hat{b}_j$$



Quantum 8, 1237 (2024).

Polaron QME: Strong coupling

A **polaron** is a **quasiparticle** consisting of a charge carrier **dressed** by the collective excitations of its surrounding medium. When a particle moves through a polarizable environment, it **distorts** that environment. The distortion then acts back on the particle, so the particle and its self-induced cloud must be treated as a **combined object**.

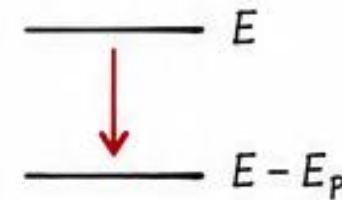


F. Chevy, Physics 9, 86 (2016)

Polaron = particle + environmental dressing

electron + nuclei in the lattice;
quantum emitters + photons or phonons.

Physics signatures



Energy renormalization

The dressed particle has a lower energy by the polaron binding energy E_p .

$$m \longrightarrow m^* > m$$

Effective mass enhancement

The particle becomes heavier due to the phonon cloud.

$$J \longrightarrow J_R = J B$$
$$0 < B < 1$$

Renormalized couplings

Hopping / tunneling amplitudes are reduced by the factor B (Franck-Condon or Debye-Waller factor).

Polaron QME: Strong coupling

Model

$$H = H_S \otimes I_B + H_{SB} + I_S \otimes H_B$$
$$\equiv (\varepsilon \sigma_z + \Delta \sigma_x) \otimes I_B + g \sigma_z \otimes \sum_k \lambda_k (b_k^\dagger + b_k) + I_S \otimes \left(\sum_k \omega_k b_k^\dagger b_k \right)$$

Unitary transformation

$$U_p = \exp \left\{ -i \sigma_z \sum_k \frac{g \lambda_k}{i \omega_k} (b_k^\dagger - b_k) \right\}$$

Mapped Hamiltonian

$$\tilde{H} = \varepsilon \sigma_z + \Delta \xi_+ \sigma_+ + \Delta \xi_- \sigma_- + H_B$$

BCH formula

$$U_p b_k U_p^\dagger = b_k - \sigma_z \frac{g \lambda_k}{\omega_k}$$
$$e^A B e^{-A} = B + [A, B] + \frac{1}{2!} [A, [A, B]]$$
$$+ \frac{1}{3!} [A, [A, [A, B]]] + \dots$$

<https://doi.org/10.1038/s42005-022-00887-2>

OPEN

Hamiltonian open quantum system toolkit

Polaron QME: Strong coupling

Mapped Hamiltonian

$$\tilde{H} = \varepsilon \sigma_z + \Delta \xi_+ \sigma_+ + \Delta \xi_- \sigma_- + H_B$$

$$\xi_{\pm} = \exp \left\{ \pm 2i \sum_k \frac{g \lambda_k}{i \omega_k} (b_k^{\dagger} - b_k) \right\}$$

$$\langle \xi_{\alpha}(t) \xi_{\beta}(0) \rangle = \text{Tr}_B \{ U_B^{\dagger}(t) \xi_{\alpha} U_B(t) \xi_{\beta} \rho_B \}$$

$$\langle \xi_{\alpha}(t) \xi_{\alpha}(0) \rangle = \kappa^2 e^{-\phi(t)}$$

$$\phi(t) = 4g^2 \int_0^{\infty} d\omega \frac{J(\omega)}{\omega^2} \left(\coth\left(\frac{\beta\omega}{2}\right) \cos(\omega t) - i \sin(\omega t) \right)$$

$$\langle \xi_{\alpha}(t) \xi_{\beta}(0) \rangle = \kappa^2 e^{\phi(t)},$$

$$\kappa = \langle \xi_{\pm}(t) \rangle = \exp \left\{ -2g^2 \sum_k \frac{\lambda_k^2}{\omega_k^2} \coth(\beta\omega_k/2) \right\} = \exp \left\{ -2g^2 \int_0^{\infty} \frac{J(\omega)}{\omega^2} \coth(\beta\omega/2) d\omega \right\};$$

Polaron QME: Strong coupling

Mapped Hamiltonian

$$\tilde{H} = \varepsilon \sigma_z + \Delta \xi_+ \sigma_+ + \Delta \xi_- \sigma_- + H_B$$

$$\xi_{\pm} = \kappa + (\xi_{\pm} - \kappa)$$

$$\sigma_+ \xi_+ + \sigma_- \xi_- = \underbrace{\kappa(\sigma_+ + \sigma_-)}_{\sigma_x} + \sigma_+(\xi_+ - \kappa) + \sigma_-(\xi_- - \kappa)$$

$$\tilde{H} = \tilde{H}_S + H_B + \tilde{H}_{SB}$$

$$\tilde{H}_S = \varepsilon \sigma_z + \kappa \Delta \sigma_x \quad \tilde{H}_{SB} = \Delta [\sigma_+(\xi_+ - \kappa) + \sigma_-(\xi_- - \kappa)]$$

$$\frac{d\tilde{\rho}_S(t)}{dt} = -\frac{1}{\hbar^2} \int_0^t ds \operatorname{Tr}_B \left[\tilde{H}_{SB}(t), \left[\tilde{H}_{SB}(s), \tilde{\rho}_S(t) \rho_B \right] \right]$$

Polaron QME: Strong coupling

PHYSICAL REVIEW LETTERS **126**, 153603 (2021)

Cavity Quantum Electrodynamics at Arbitrary Light-Matter Coupling Strengths

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Quantum light-matter systems at strong coupling are notoriously challenging to analyze due to the need to include states with many excitations in every coupled mode. We propose a nonperturbative approach to analyze light-matter correlations at all interaction strengths. The key element of our approach is a unitary transformation that achieves asymptotic decoupling of light and matter degrees of freedom in the limit where light-matter interaction becomes the dominant energy scale. In the transformed frame, truncation of the matter or photon Hilbert space is increasingly well justified at larger coupling, enabling one to systematically derive low-energy effective models, such as tight-binding Hamiltonians. We demonstrate the versatility of our approach by applying it to concrete models relevant to electrons in crystal potential and electric dipoles interacting with a cavity mode. A generalization to the case of spatially varying electromagnetic modes is also discussed.

Polaron QME: Strong coupling

PHYSICAL REVIEW LETTERS **126**, 153603 (2021)

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$$\hat{H}_C = \int dx \hat{\psi}_x^\dagger \left[\frac{(-i\hbar\partial_x - q\hat{A})^2}{2m} + V(x) \right] \hat{\psi}_x + \hbar\omega_c \hat{a}^\dagger \hat{a} + \hat{H}_\parallel$$

$$\begin{aligned} \hat{H}_C = \int dx \hat{\psi}_x^\dagger \left[-\frac{\hbar^2 \partial_x^2}{2m} + V(x) \right] \hat{\psi}_x + \hbar\Omega \hat{b}^\dagger \hat{b} + \hat{H}_\parallel \\ - gx_\Omega \int dx \hat{\psi}_x^\dagger (-i\hbar\partial_x) \hat{\psi}_x (\hat{b} + \hat{b}^\dagger), \end{aligned}$$

To asymptotically decouple light and matter degrees of freedom, we propose to use a unitary transformation

$$\hat{U} = \exp \left[-i\xi_g \int_{-\infty}^{\infty} dx \hat{\psi}_x^\dagger (-i\partial_x) \hat{\psi}_x \hat{\pi} \right], \quad (3)$$

Polaron QME: Strong coupling

Benchmarking the Exponential Ansatz for the Holstein model

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[arXiv:2405.18771v1](https://arxiv.org/abs/2405.18771v1)

Polarons are quasiparticles formed as a result of lattice distortions induced by charge carriers. The single-electron Holstein model captures the fundamentals of single polaron physics. We examine the power of the exponential ansatz for the polaron ground-state wavefunction in its coupled cluster, canonical transformation, and (canonically transformed) perturbative variants across the parameter space of the Holstein model. Our benchmark serves to guide future developments of polaron wavefunctions beyond the single-electron Holstein model.

Holstein Lattice

$$H = -t \sum_{\langle lk \rangle} \left(a_l^\dagger a_k + a_k^\dagger a_l \right) + \omega \sum_l b_l^\dagger b_l + g \sum_l \rho_l \left(b_l^\dagger + b_l \right)$$

$$\begin{aligned} |\Phi_0\rangle &= |\Phi_0^F\rangle \otimes |\Phi_0^B\rangle \\ |\Phi_0^F\rangle &= \sum_l A_l a_l^\dagger |0^F\rangle \\ |\Phi_0^B\rangle &= \exp \left[\sum_l \xi_l b_l^\dagger - \xi_l^* b_l \right] |0^B\rangle \\ |\Psi_{CC}\rangle &= \exp(T_{CC}) |\Phi_0\rangle \end{aligned}$$

Outline: Today's talk

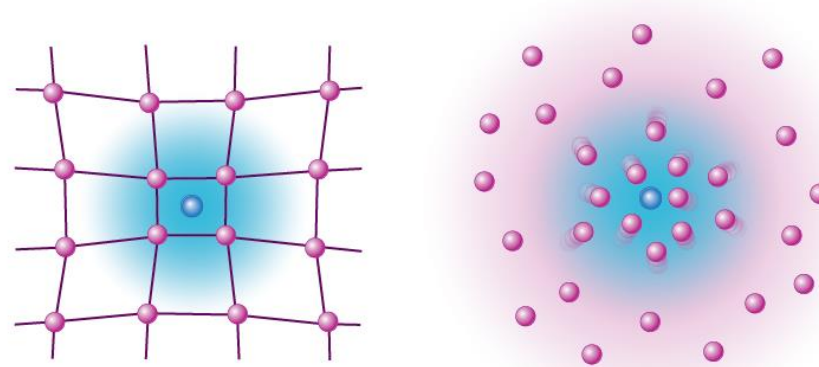
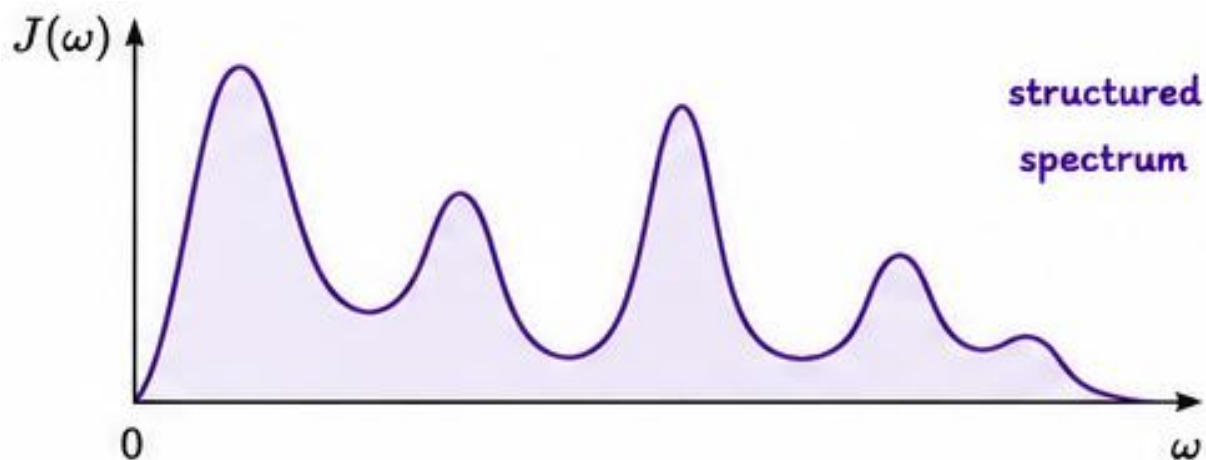
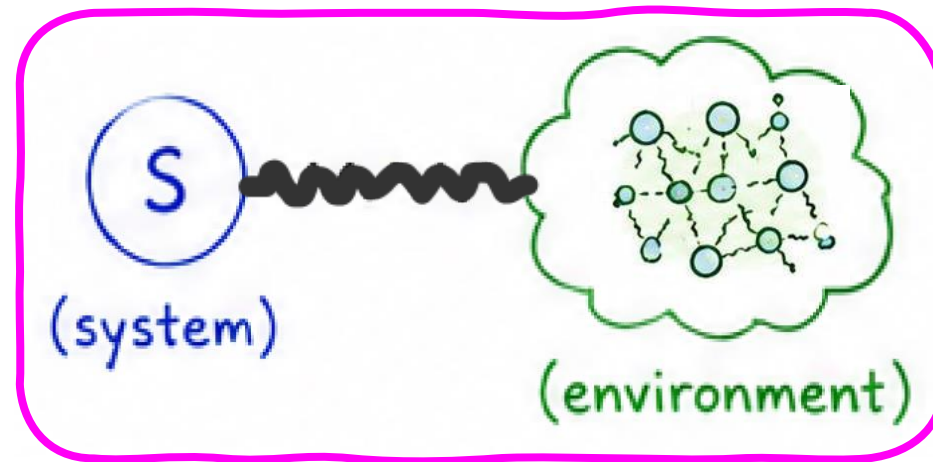
○ Review

- Weak coupling, Markovian approximation
- Quantum master equation

○ Memory effects: Measures, Methods

○ Strong coupling: Polaron mapping

○ Other methods: Repeated interaction, Path integral, HEOM,...

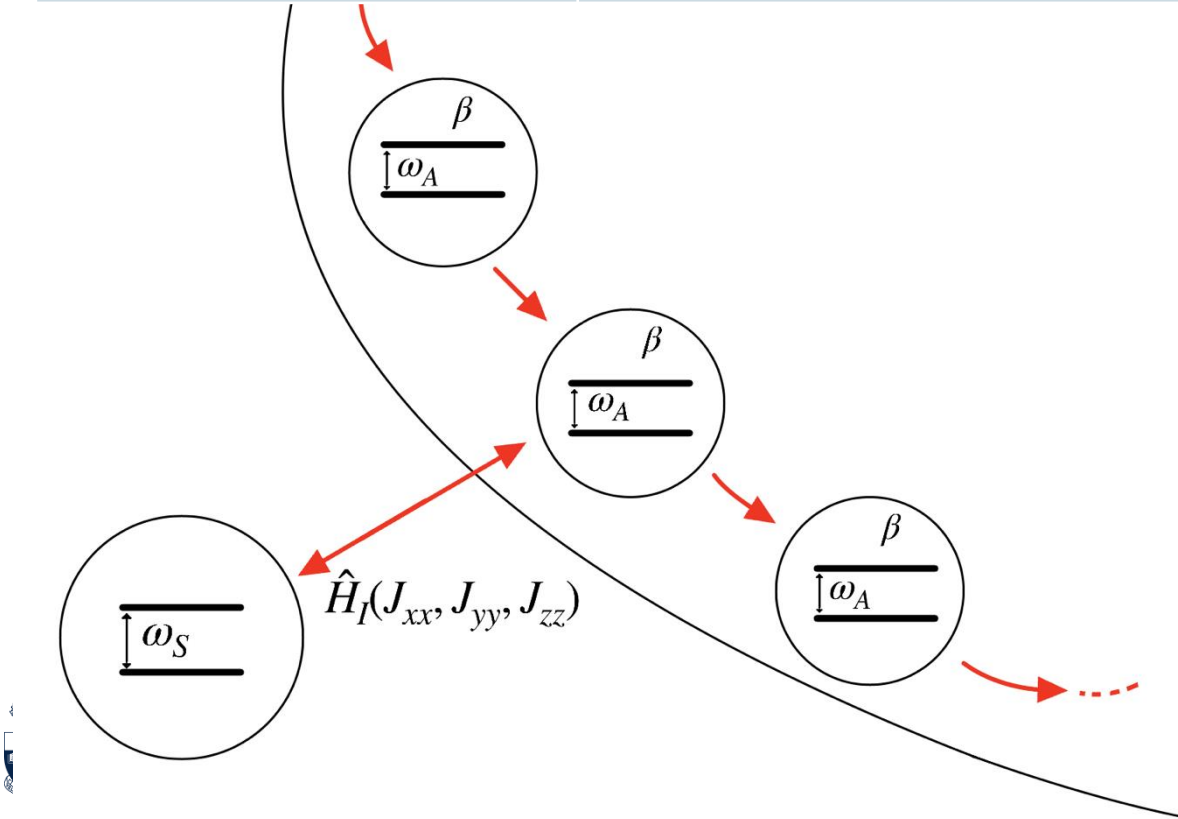


Theory of Open Quantum systems: Repeated Interaction protocol

Research Field	Question	Method	Ref
Quantum Information Science, Open Q systems	Fast, accurate and efficient state preparation on a quantum computer?	Repeated Interaction protocol	Quantum Science and Technology 10, 025061 (2025) Phys. Rev. Research, 7 (4), 043012 (2025).

Equilibrium and nonequilibrium steady states with the repeated interaction protocol: Relaxation dynamics and energetic cost
Alessandro Porssitto, Madeline Forbes, Dvira Segal
[Quantum Science & Technology 10, 025061 \(2025\)](#)

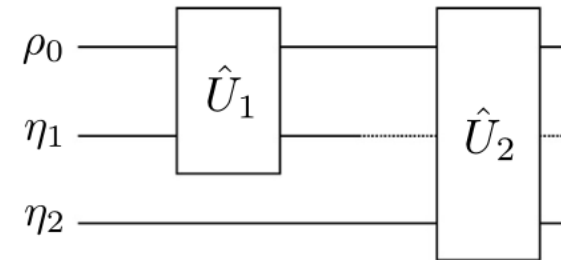
Thermal state preparation by repeated interactions at and beyond the Lindblad limit,
Carlos Ramon-Escandell, Alessandro Prositto, D. Segal
[Phys. Rev. Research, 7 \(4\), 043012 \(2025\).](#)



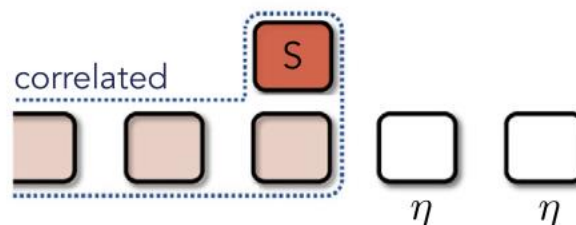
Repeated interaction (collision) models

1. **Open quantum systems:** understand the role of strong coupling, non Markovianity; achieve dynamics beyond the Lindblad equation
2. **Physics:** Thermalization, nonequilibrium steady state, thermal machines
3. **Quantum algorithms:** Emulate open system dynamics with unitary dynamics and measurements (quantum algorithms)
4. **Physical systems**

Markovian scheme



Non Markovian scheme



Quantum collision models: Open system dynamics from repeated interactions.” Physics Reports 954 (2022): 1-70.

A Digital Quantum Algorithm for Non-Markovian Electron Transfer Dynamics Using Repeated Interactions

Lea K. Northcote,^{1,*)} Matthew S. Teynor,^{1,2,*)} and Gemma C. Solomon^{1,2,†)}

arXiv:2502.18426v2

Efficient quantum algorithm to simulate open systems through a single environmental qubit

Giovanni Di Bartolomeo ,^{1,2,*,†} Michele Vischi ,^{1,2,*,‡} Tommaso Feri ,^{1,2,*,§} Angelo Bassi ,^{1,2} and Sandro Donadi³

Phys. Rev. Research 6, 043321(2024)

Collision Models Can Efficiently Simulate Any Multipartite Markovian Quantum Dynamics

[Marco Cattaneo](#), [Gabriele De Chiara](#), [Sabrina Maniscalco](#), [Roberta Zambrini](#), and [Gian Luca Giorgi](#)

Phys. Rev. Lett. **126**, 130403 (2021)

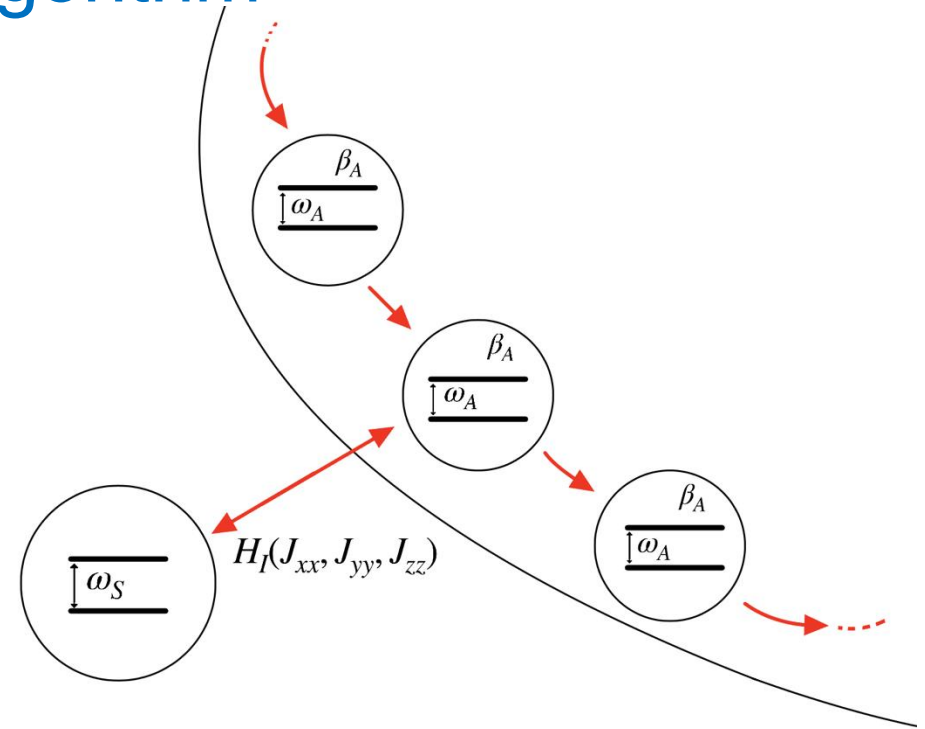
Thermalization of a system

Repeated interaction as a quantum algorithm

$$\hat{H}_S = -\frac{\omega_S}{2}\hat{\sigma}_z^S, \quad \hat{H}_A = -\frac{\omega_A}{2}\hat{\sigma}_z^A$$

$$\hat{H}_{SA} = J_{xx}\hat{\sigma}_x^S \otimes \hat{\sigma}_x^A + J_{yy}\hat{\sigma}_y^S \otimes \hat{\sigma}_y^A$$

$$\begin{aligned} \boxed{\rho_S^{(n+1)}} &= \text{Tr}_A \left[\rho_{tot}^{(n+1)} \right] \\ &= \text{Tr}_A \left[\hat{U}(\tau) \rho_{tot}^{(n)} \hat{U}^\dagger(\tau) \right] \\ &= \text{Tr}_A \left[\hat{U}(\tau) \left(\rho_S^{(n)} \otimes \rho_A \right) \hat{U}^\dagger(\tau) \right] \end{aligned}$$

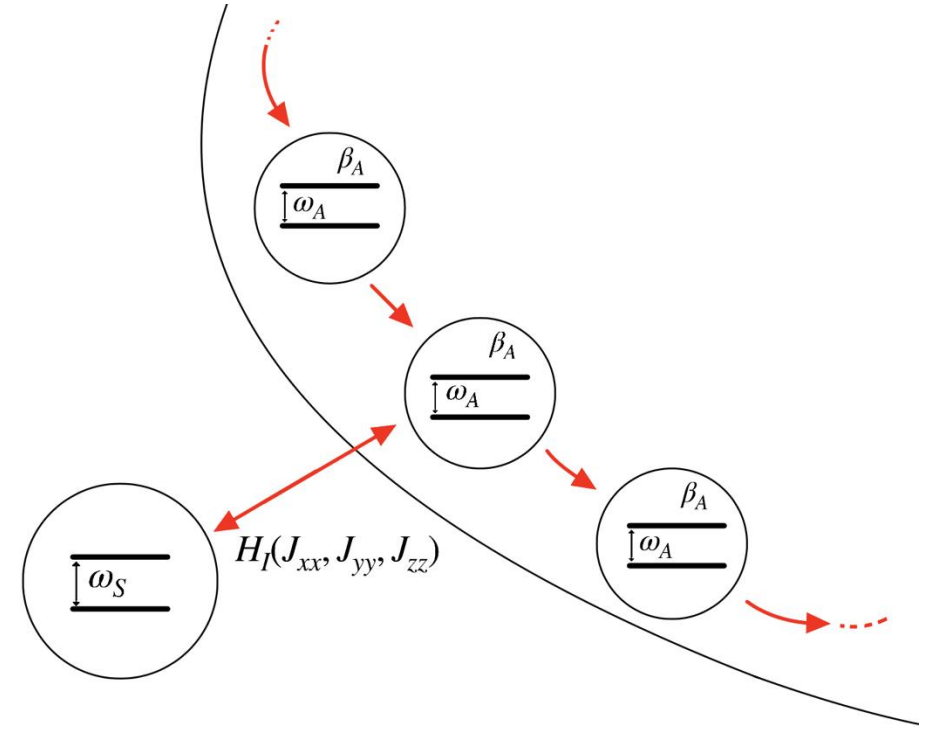


Thermalization of a system on a quantum computer: Repeated interaction

$$p_S^{(n)} - p_S^{(\infty)} = \eta^n \left(p_S^{(0)} - p_S^{(\infty)} \right)$$

$$\eta = 1 - \frac{4(J_{xx} + J_{yy})^2}{\theta^2} \sin^2\left(\frac{\theta\tau}{2}\right) - \frac{4(J_{xx} - J_{yy})^2}{\phi^2} \sin^2\left(\frac{\phi\tau}{2}\right)$$

$$\frac{dp_1(t)}{dt} = -k_u p_1 + k_d p_2, \quad \frac{dp_2(t)}{dt} = -k_d p_2 + k_u p_1$$

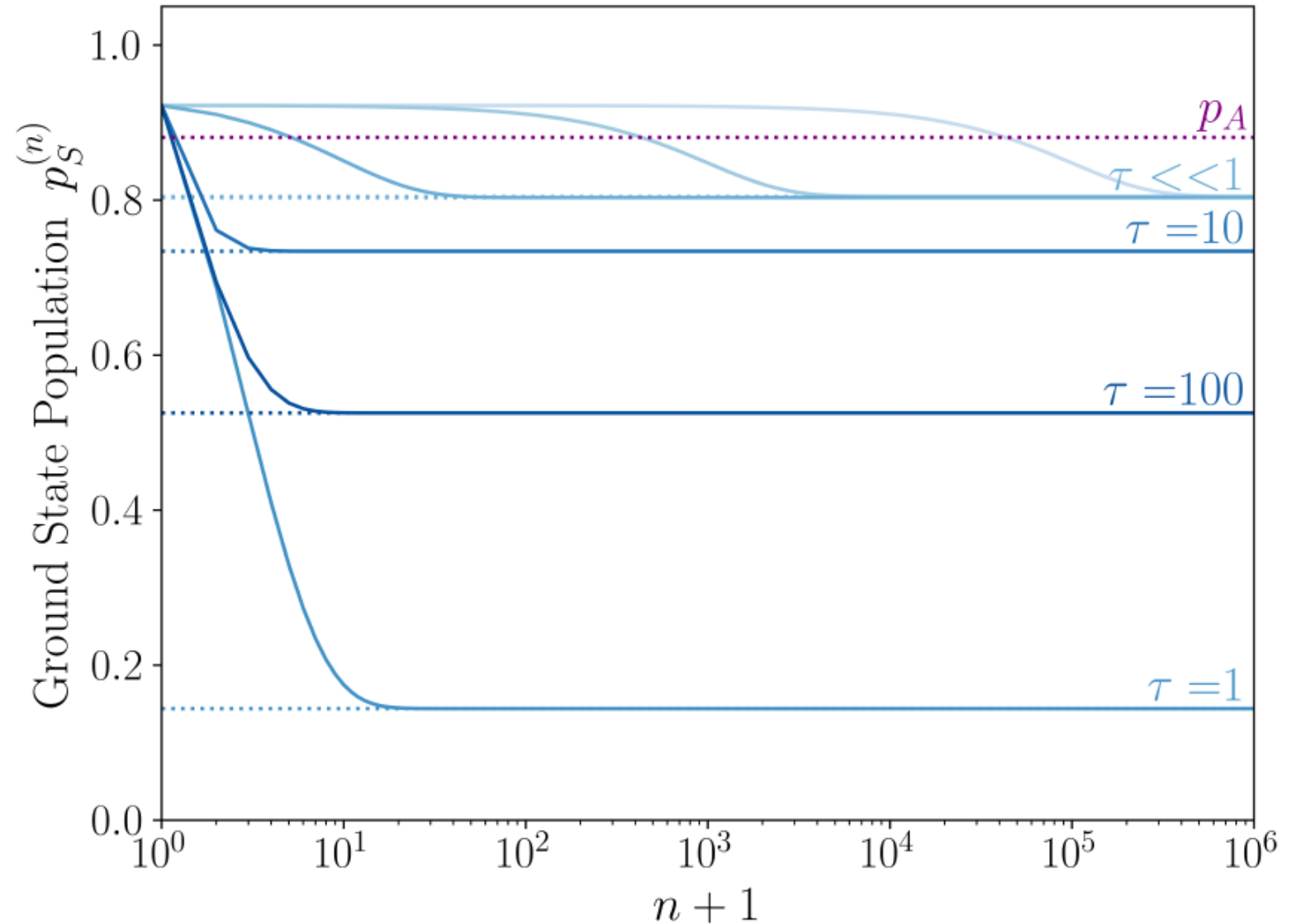
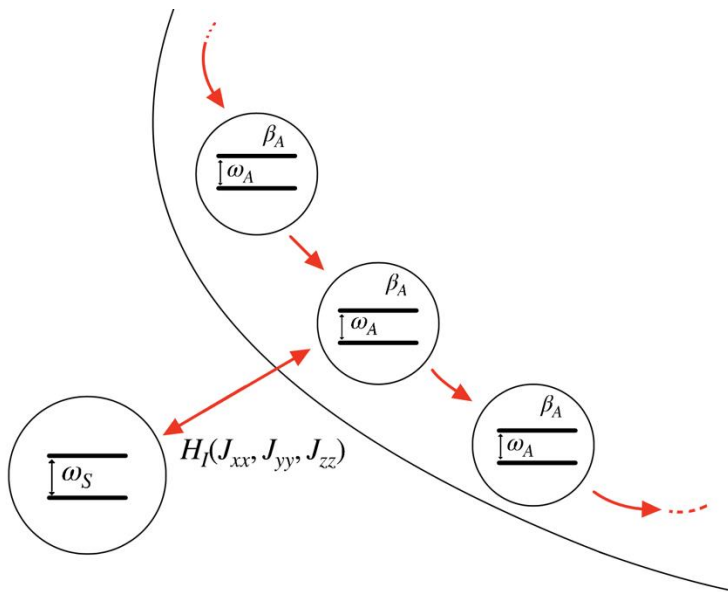


$$k_d = 4J_{xx}J_{yy}\tau \frac{e^{\beta\omega_A}}{e^{\beta\omega_A} + 1} + (J_{xx} - J_{yy})^2 \tau,$$

$$k_u = -4J_{xx}J_{yy}\tau \frac{e^{\beta\omega_A}}{e^{\beta\omega_A} + 1} + (J_{xx} + J_{yy})^2 \tau.$$

Thermalization of a system on a quantum computer: Repeated interaction

$$p_S^{(n)} - p_S^{(\infty)} = \eta^n (p_S^{(0)} - p_S^{(\infty)})$$



Thermal state preparation

Many collisions after...

Initial condition:
completely mixed state

target state: Lower temperature

Thermal state preparation: Mpemba effect

$$D(\rho_S(T_{\text{sim}}), \rho_S^*) = \frac{1}{2} \text{Tr} |\rho_S(T_{\text{sim}}) - \rho_S^*| \leq \epsilon$$

